

# **Lecture 3:**

# **Image Transformation**

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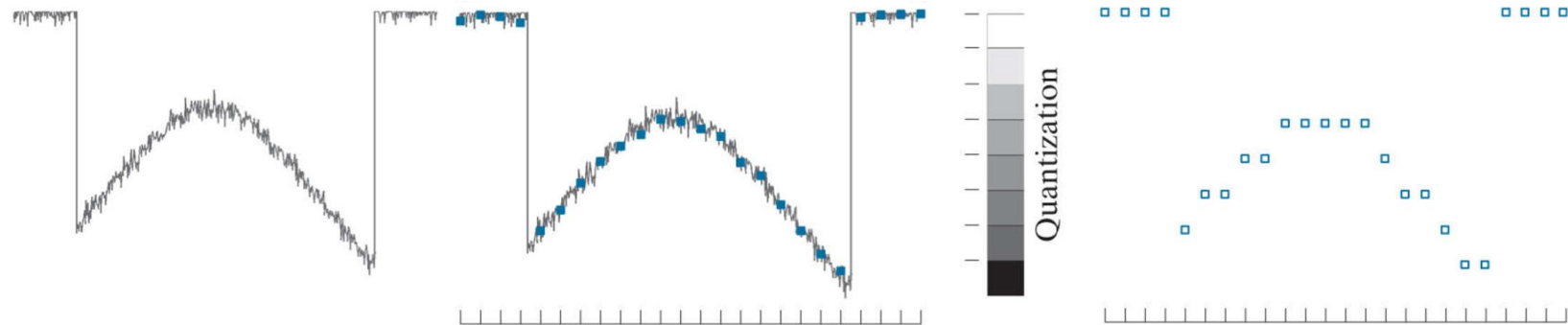
# How was you lab 1?



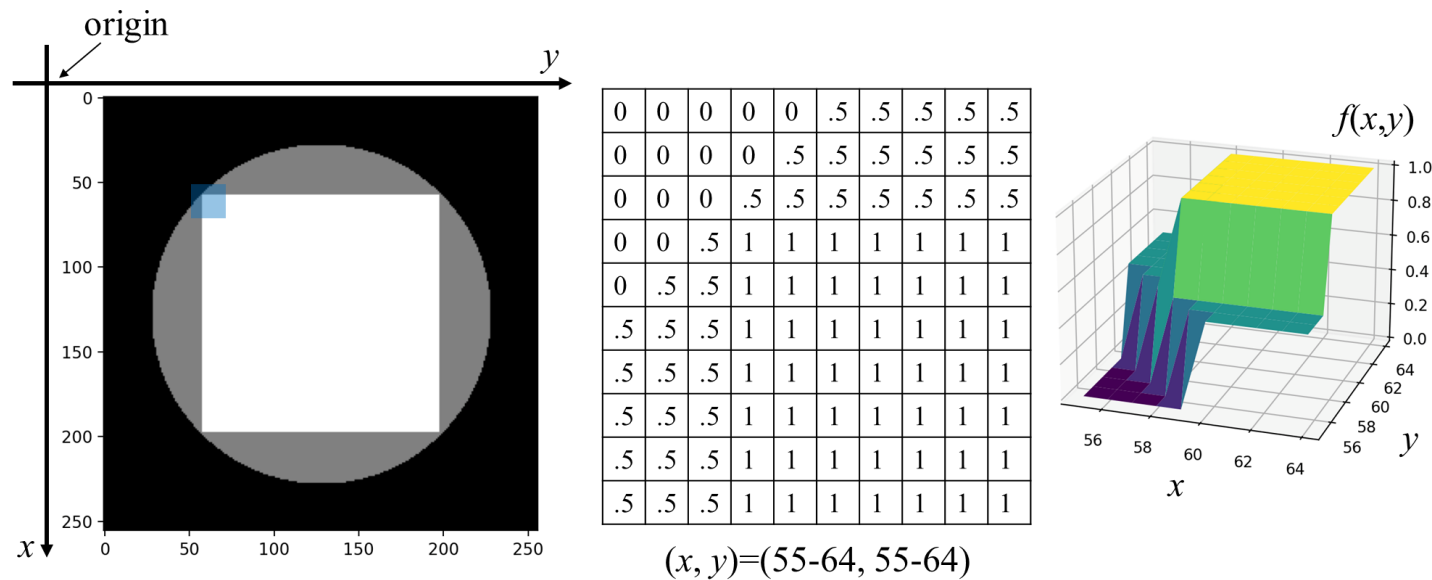
**Do you prefer AI-based coding,  
or the traditional way?**

# Previously ...

- Digitalize



- Matrix



# Outline

- Mathematical operations
  - Basic, Logical operations
- Spatial transform
  - Rigid, affine, nonlinear
- Intensity Transform
  - Point transform (pixel-wise)
  - Histogram (probability)
  - Spatial filter

# Basic Mathematical Operations

Pixelwise operator

Arithmetic operator

Set and logic operator

# Elementwise

- Not only matrix but **spatial** information

- Addition

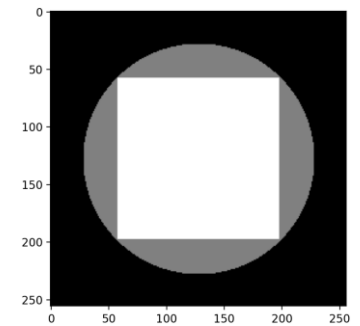
$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix}$$

- Matrix production

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

- **Pixel-wise** production

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \odot \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$



# Linear operation

If H is a linear operator: **additivity** and **homogeneity**

$$H\left(af_1(x, y) + bf_2(x, y)\right) = aH\left(f_1(x, y)\right) + bH\left(f_2(x, y)\right)$$

Give an example of H?

- (A) H is a shift operator
- (B) H is an absolute value operator
- (C) H is a Fourier operator

# Arithmetic operation

- Pixel-wised: addition

$$s(x, y) = f_1(x, y) + f_2(x, y)$$

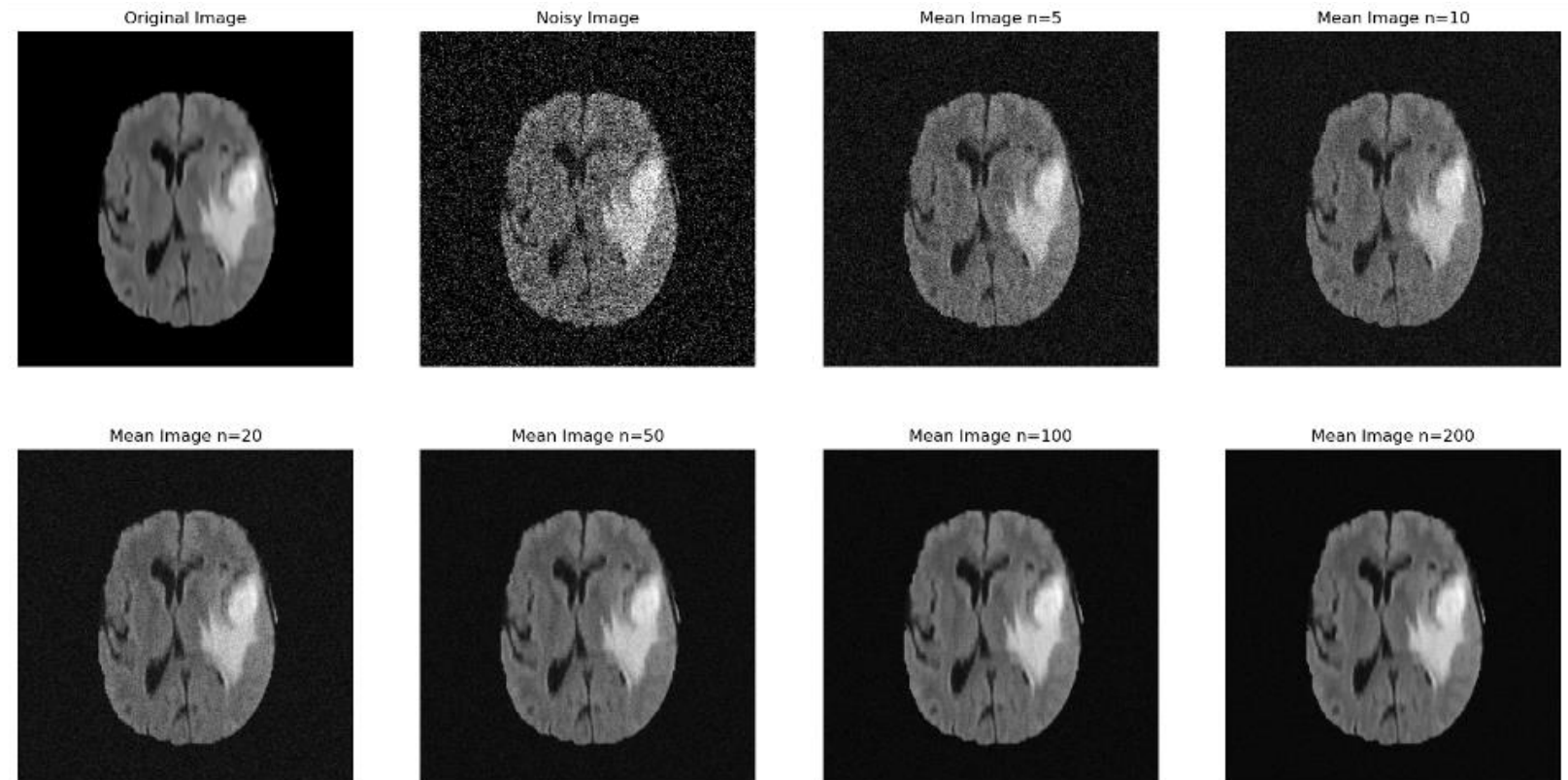
Reduce noise:

$$g(x, y) = f(x, y) + \eta(x, y)$$

$$\bar{g}(x, y) = \frac{1}{K} \sum_{i=1}^K g_i(x, y)$$

$$E\{\bar{g}(x, y)\} = f(x, y)$$

$$\sigma_{\bar{g}(x, y)}^2 = \frac{1}{K} \sigma_{\eta(x, y)}^2$$





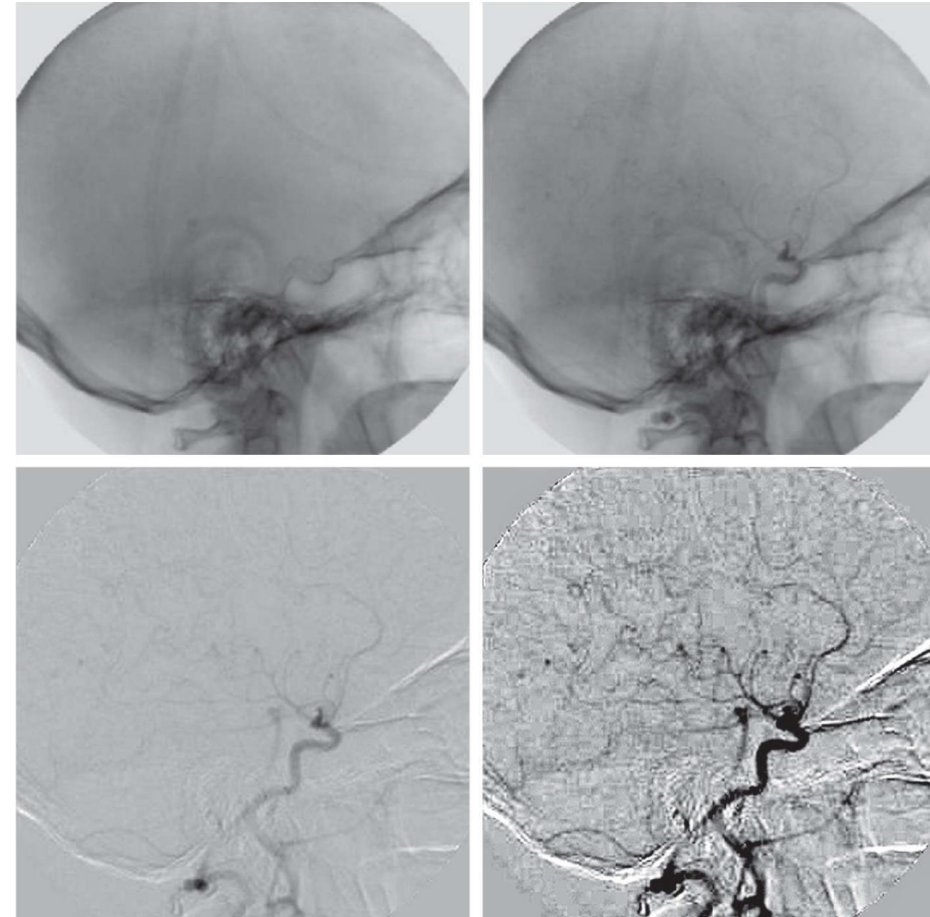
# Arithmetic operation

- Pixel-wised: subtraction

$$d(x, y) = f_1(x, y) - f_2(x, y)$$

→ Find differences

## Digital Subtraction Angiography (DSA)



difference

enhanced

# Arithmetic

- Pixel-wised: multiplication and division

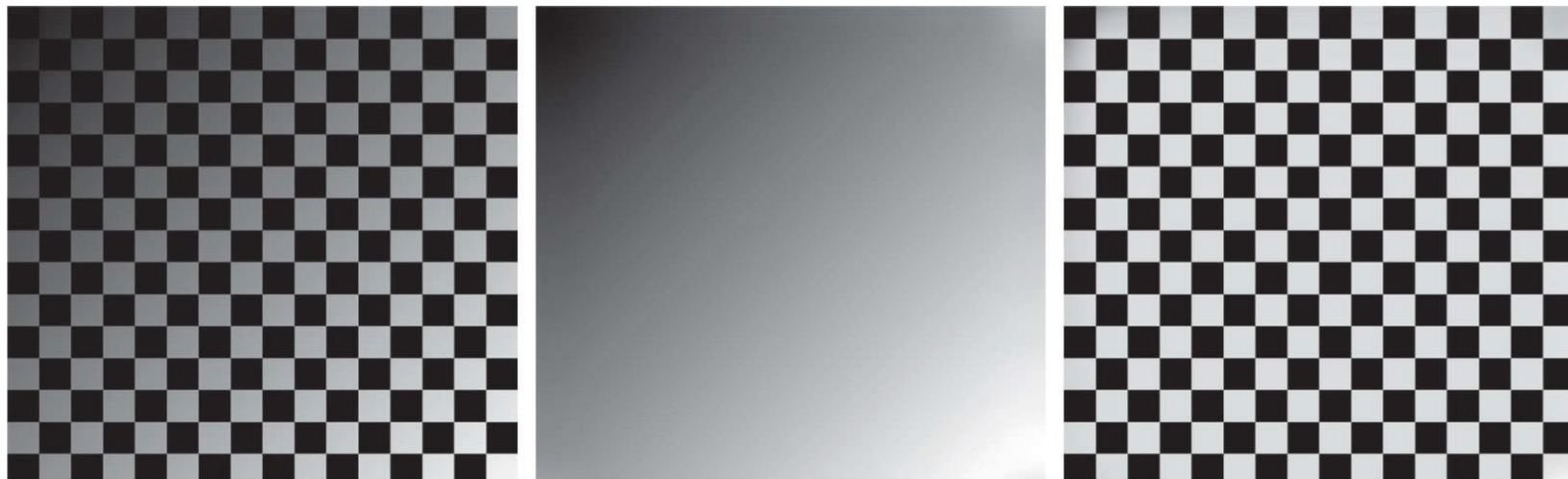
$$p(x, y) = f_1(x, y) \times f_2(x, y)$$

Mask region



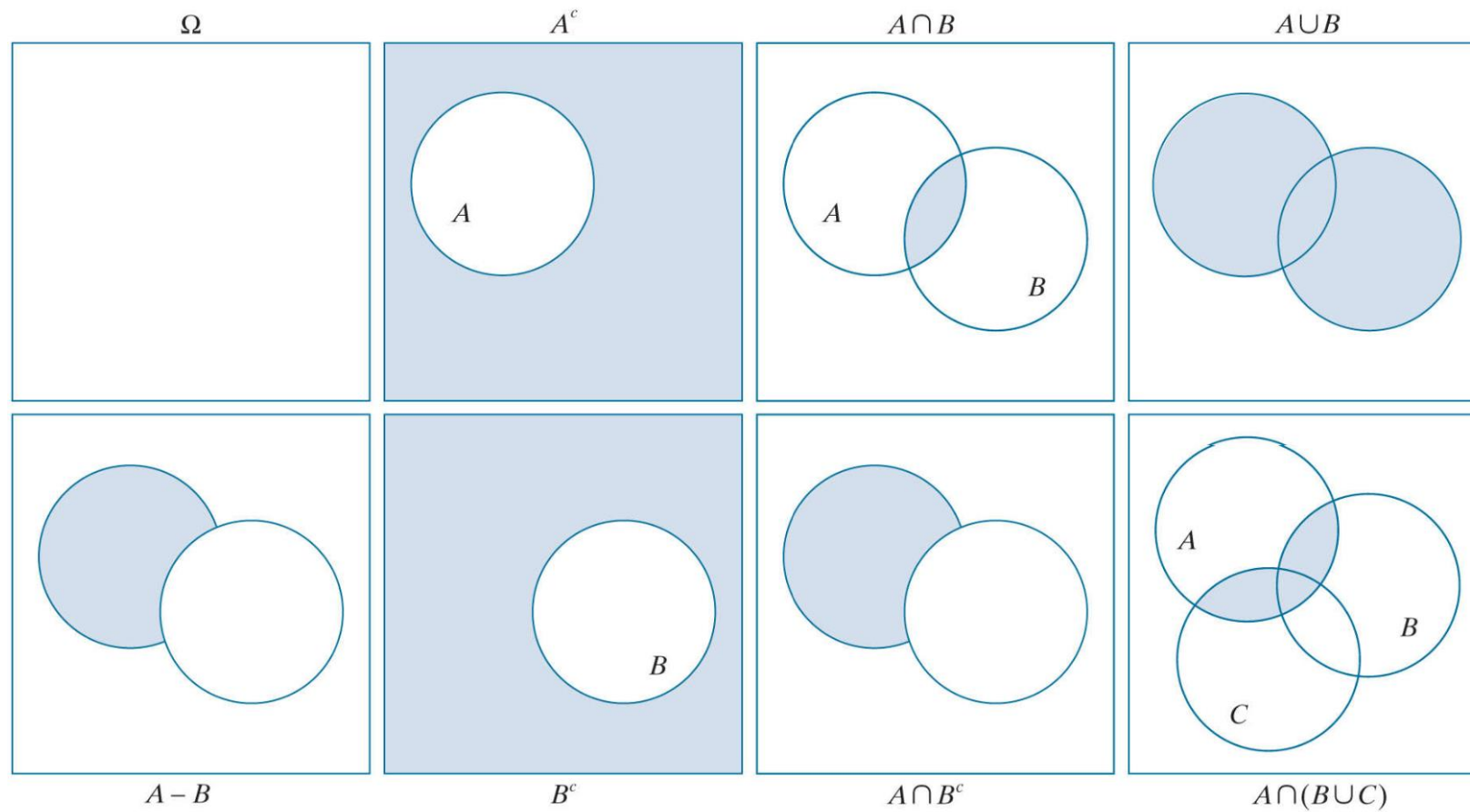
$$v(x, y) = f_1(x, y) \div f_2(x, y)$$

Normalize

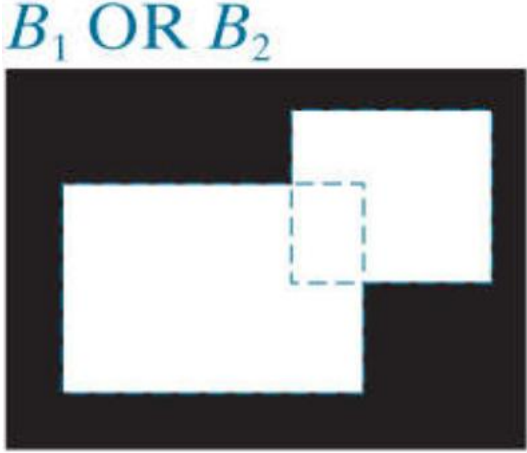
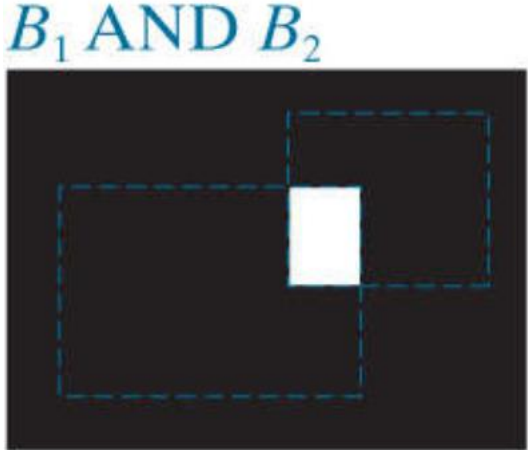
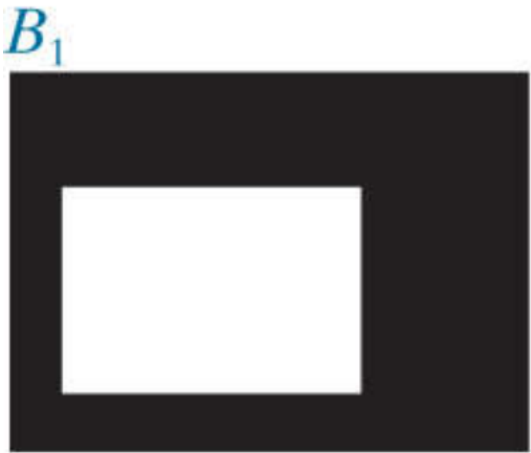


# Set operations

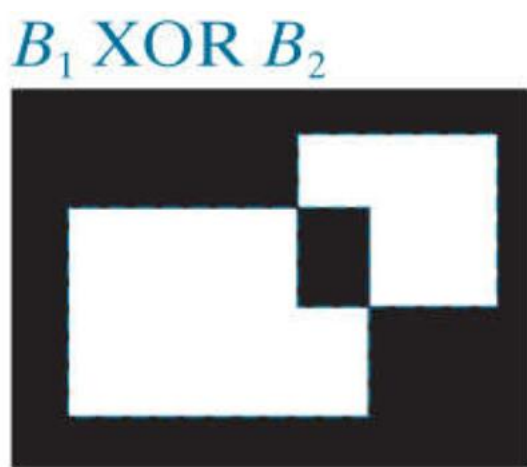
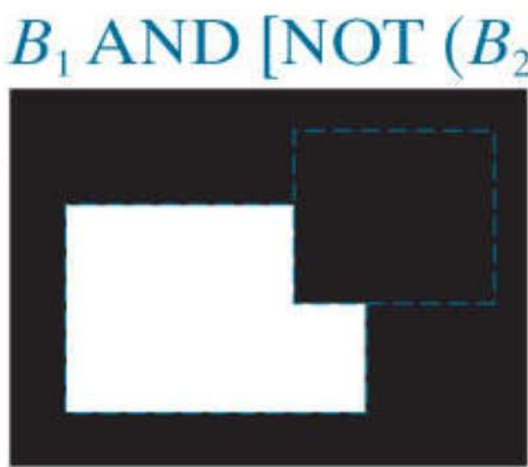
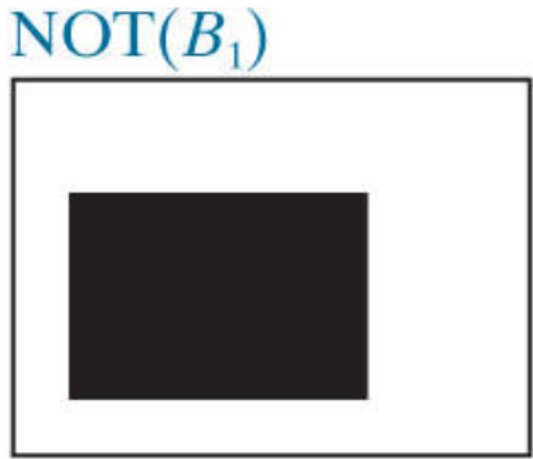
- Binary image



# Logical operations



|     |   |   |   |   |
|-----|---|---|---|---|
| B1  | 1 | 1 | 0 | 0 |
| B2  | 1 | 0 | 1 | 0 |
| not | 0 | 0 | 1 | 1 |
| and | 1 | 0 | 0 | 0 |
| or  | 1 | 1 | 1 | 1 |
| xor | 0 | 1 | 1 | 0 |



# Spatial Transformation

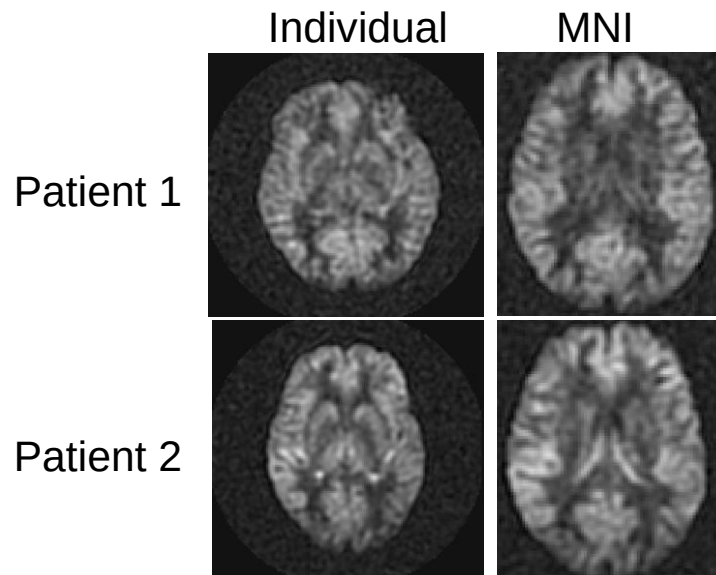
Rigid transform

Affine transform

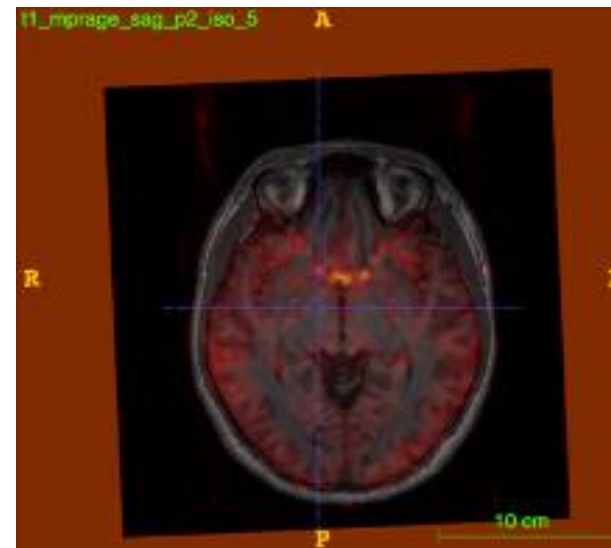
Non-linear transform

# Spatial Transform

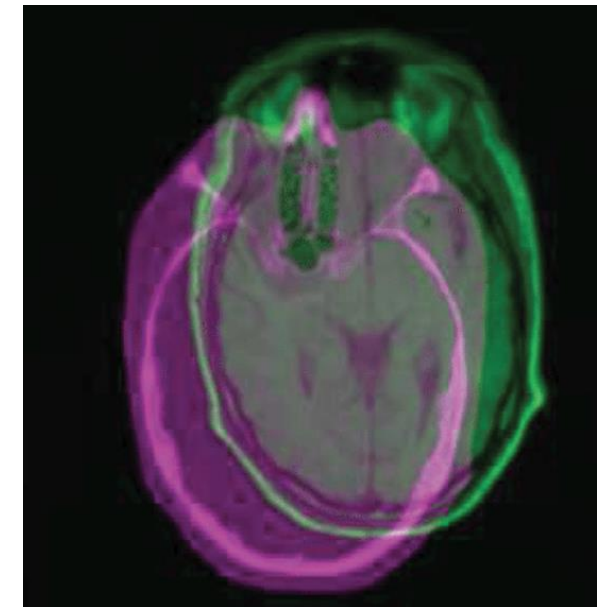
- Why we need transformation?
  - Registration



Comparison of different individuals



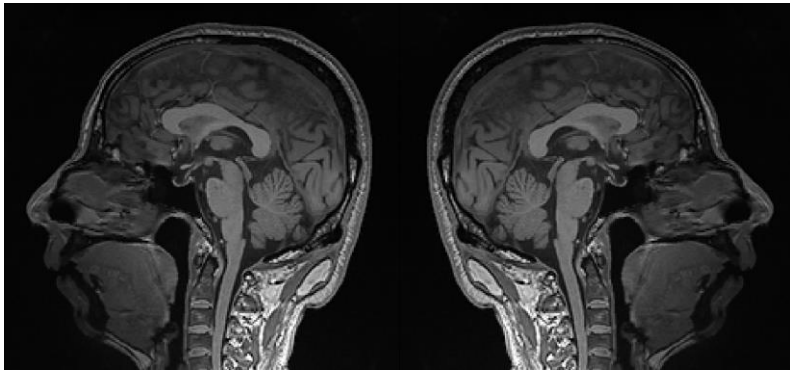
ASL→T1



Radiotherapy planning  
using CT and MRI

# Spatial Transform

Image Flip



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \mathbf{T} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

If  $x \in [-N/2, N/2]$

$$T = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Usually  $x \in [0, N - 1]$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 0 & N \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



# Spatial Transform

$$A = \begin{bmatrix} \overset{\text{scale}}{\boxed{a_{11}}} & \overset{\text{rotation}}{\boxed{a_{12}}} & \overset{\text{translation}}{\boxed{a_{13}}} \\ \boxed{a_{21}} & \boxed{a_{22}} & \boxed{a_{23}} \\ \underset{\text{shear}}{0} & 0 & 1 \end{bmatrix}$$

## Affine transformations

- Translation
  - Rotation
  - Scaling
  - Shearing
- $\left. \begin{array}{l} \text{Translation} \\ \text{Rotation} \end{array} \right\} \text{Rigid} \left\{ \begin{array}{l} \text{9-degree} \\ \text{Affine} \end{array} \right\} \left\{ \begin{array}{l} \text{12-degree} \\ \text{Affine} \end{array} \right\}$



# Spatial Transform

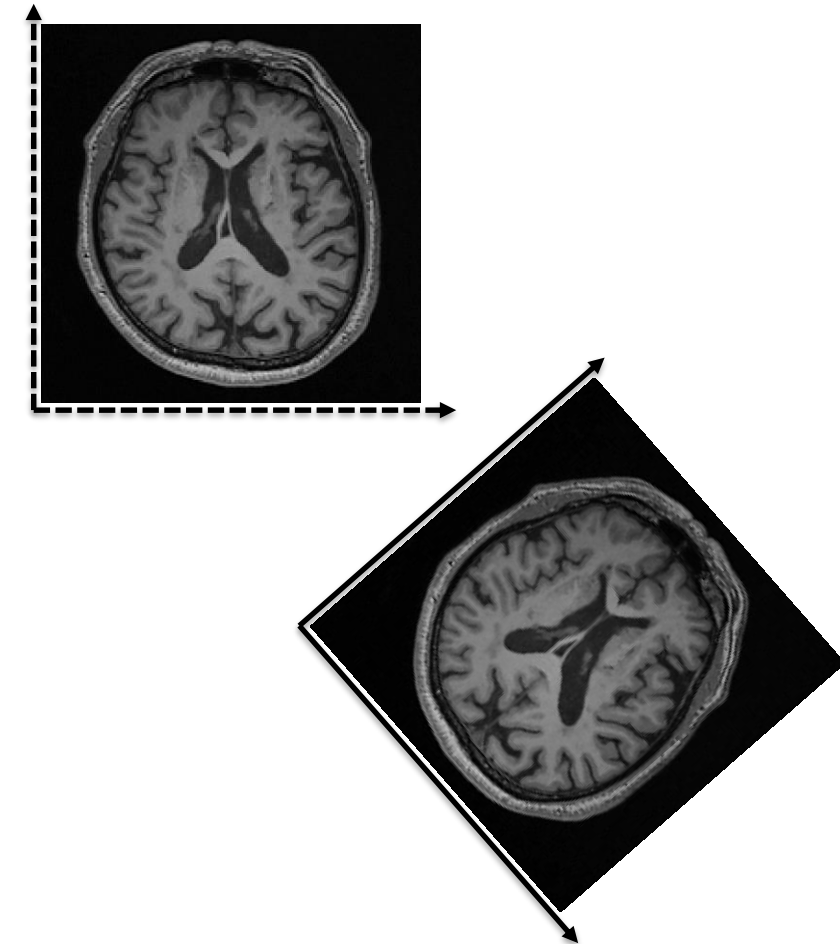
- **Rigid-body Transform**

- Translation
- Rotation

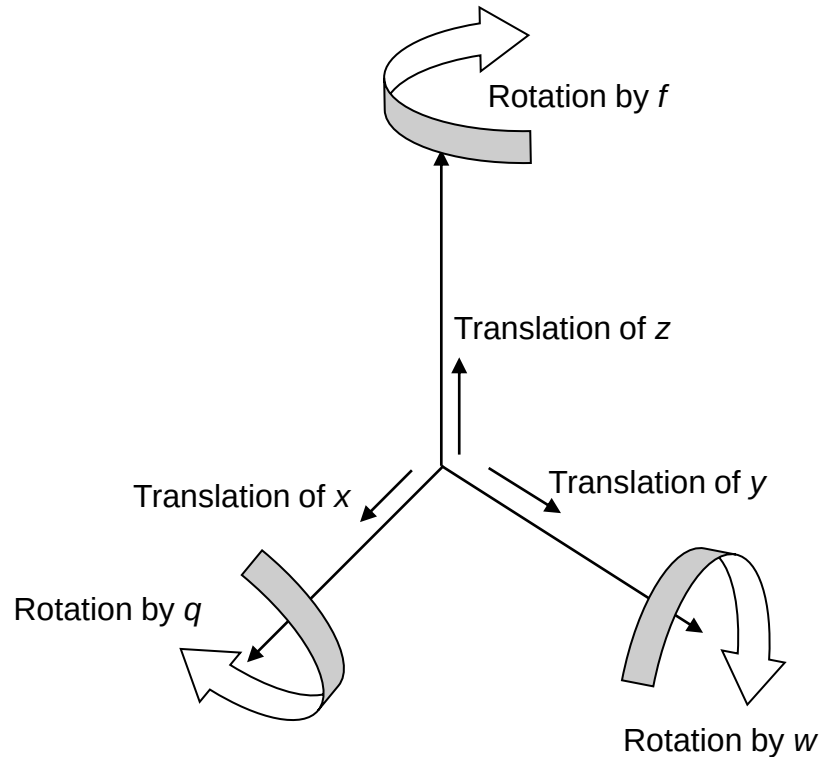
$$A = \begin{bmatrix} \begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{matrix} & \begin{matrix} a_{13} \\ a_{23} \end{matrix} \\ 0 & 0 & 1 \end{bmatrix}$$

rotation                  translation

$$R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$



# 3-D Rigid Transform



$$x' = Rx + t$$

Translation vector

- Translation along x-axis by  $p$

$$x' = x + p$$

$$y' = y$$

$$z' = z$$

- Translation along y-axis by  $q$

$$x' = x$$

$$y' = y + q$$

$$z' = z$$

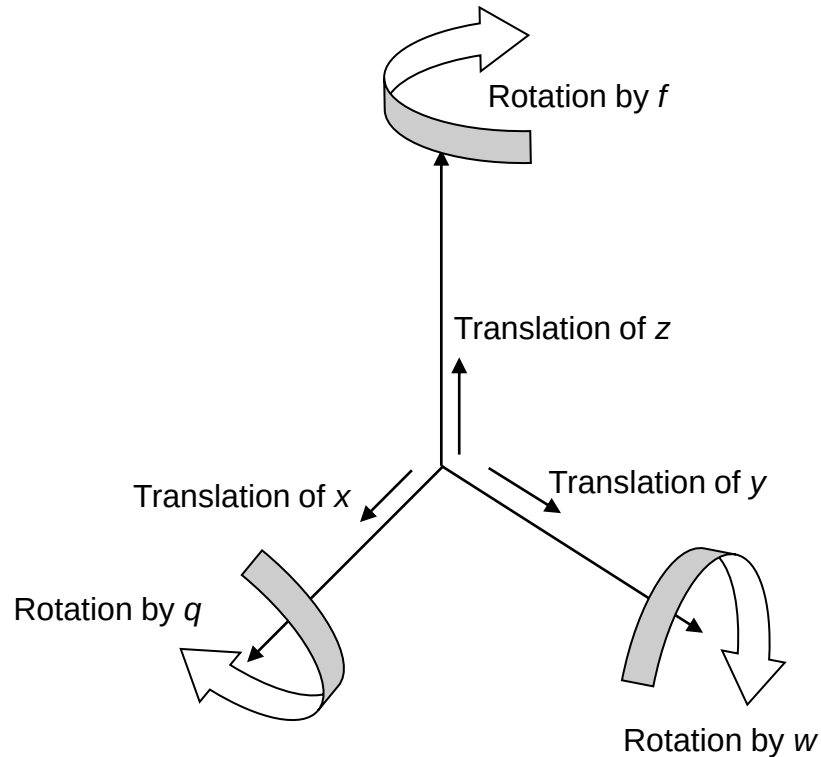
- Translation along z-axis by  $r$

$$x' = x$$

$$y' = y$$

$$z' = z + r$$

# 3-D Rigid Transform



$$x' = Rx + t$$

↗ ↘  
**Rotation matrix**      **Translation vector**

- Rotation about x-axis by  $q$

$$x' = x$$

$$y' = y \cos \theta + z \sin \theta$$

$$z' = -y \sin \theta + z \cos \theta$$

- Rotation about y-axis by  $w$

$$x' = x \cos \omega - z \sin \omega$$

$$y' = y$$

$$z' = x \sin \omega + z \cos \omega$$

- Rotation about z-axis by  $\phi$

$$x' = x \cos \phi + y \sin \phi$$

$$y' = -x \sin \phi + y \cos \phi$$

$$z' = z$$

## 3-D Rigid Transform

- A 3D rigid body transform is defined by:
  - 3 translations - in X, Y & Z directions
  - 3 rotations - about X, Y & Z axes
- Does the order of operations matters?

$$R = R_{\theta} R_{\omega} R_{\phi} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \omega & 0 & -\sin \omega \\ 0 & 1 & 0 \\ \sin \omega & 0 & \cos \omega \end{bmatrix} \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Pitch  
about x axis

Roll  
about y axis

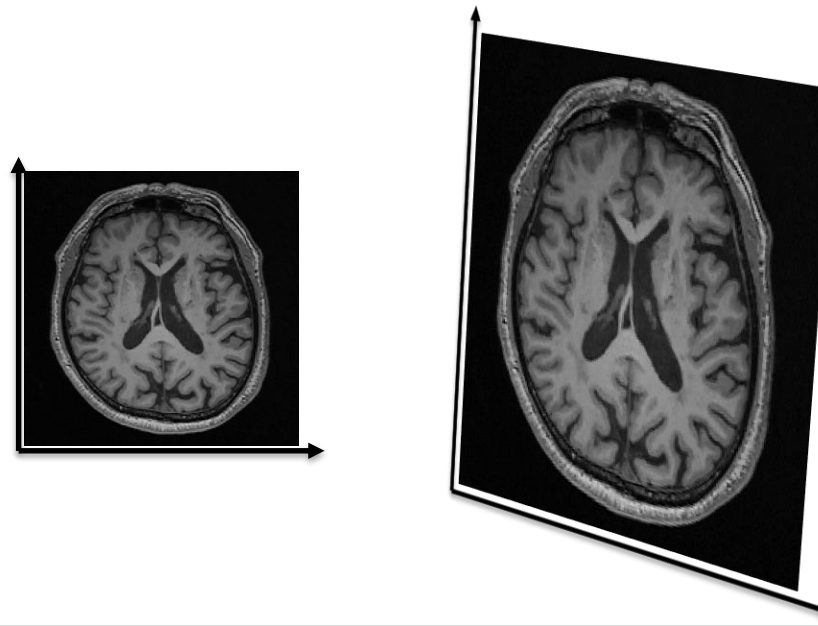
Yaw  
about z axis

# Spatial Transform

- **Affine Transform**

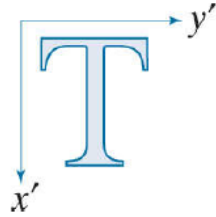
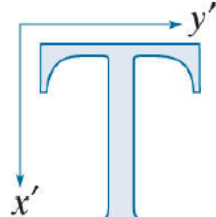
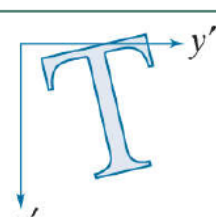
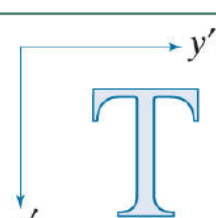
- Translation
- Rotation
- Scaling
- shearing

$$A = \begin{bmatrix} \overset{\text{scale}}{\boxed{a_{11}}} & \boxed{a_{12}} & a_{13} \\ \boxed{a_{21}} & \boxed{a_{22}} & a_{23} \\ \underset{\text{shear}}{0} & 0 & 1 \end{bmatrix}$$



# Spatial Transform

TABLE 2.3  
Affine transformations based on Eq. (2-45)

| Transformation Name                                                                  | Affine Matrix, A                                                                                         | Coordinate Equations                                                     | Example                                                                               |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Identity                                                                             | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$                                      | $x' = x$<br>$y' = y$                                                     |    |
| Scaling/Reflection (For reflection, set one scaling factor to -1 and the other to 0) | $\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$                                  | $x' = c_x x$<br>$y' = c_y y$                                             |    |
| Rotation (about the origin)                                                          | $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | $x' = x \cos\theta - y \sin\theta$<br>$y' = x \sin\theta + y \cos\theta$ |   |
| Translation                                                                          | $\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$                                  | $x' = x + t_x$<br>$y' = y + t_y$                                         |  |

# 3-D Affine Transform

$$x' = Ax$$

translation, rotation and scaling  
transformation with nine parameters

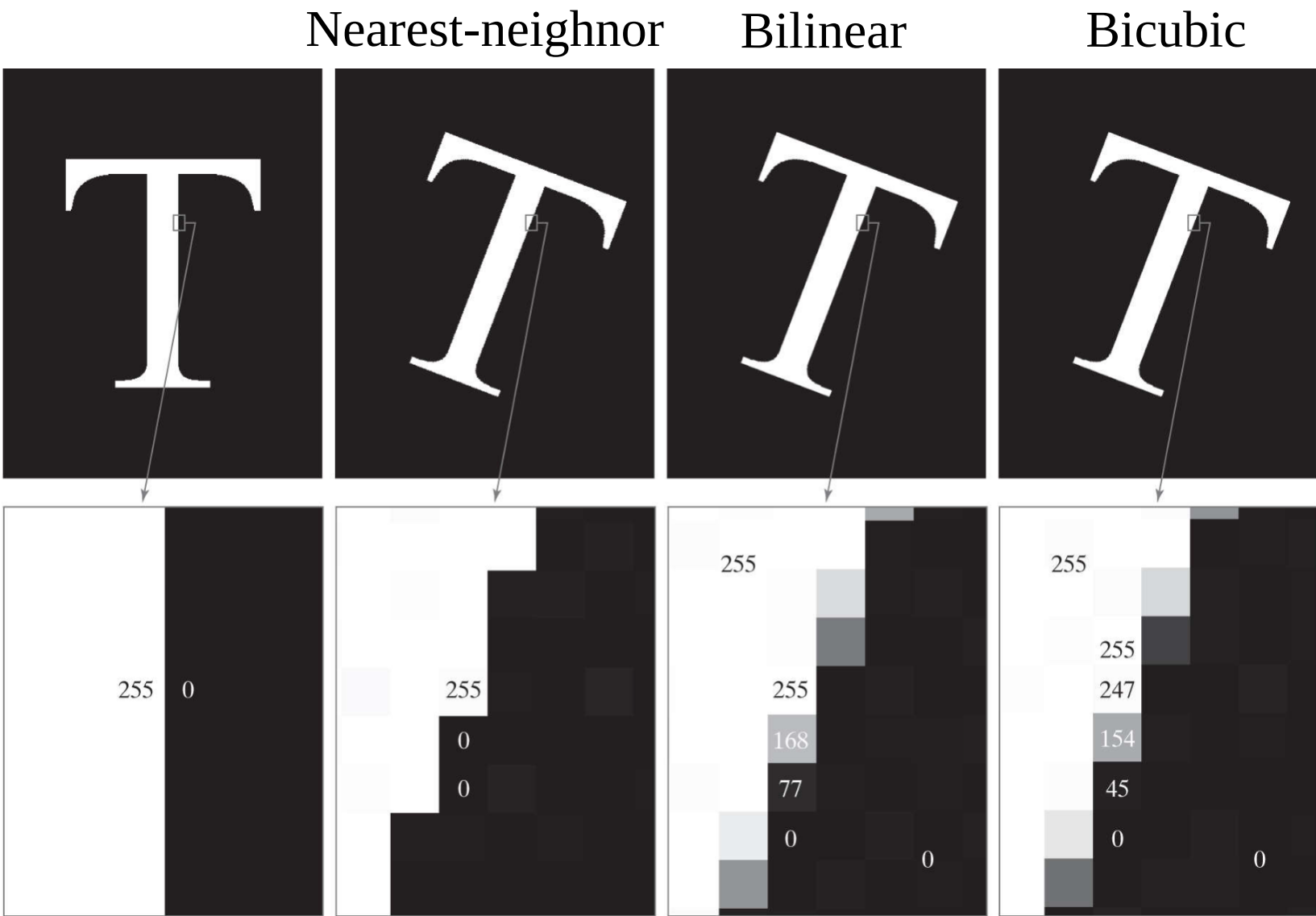
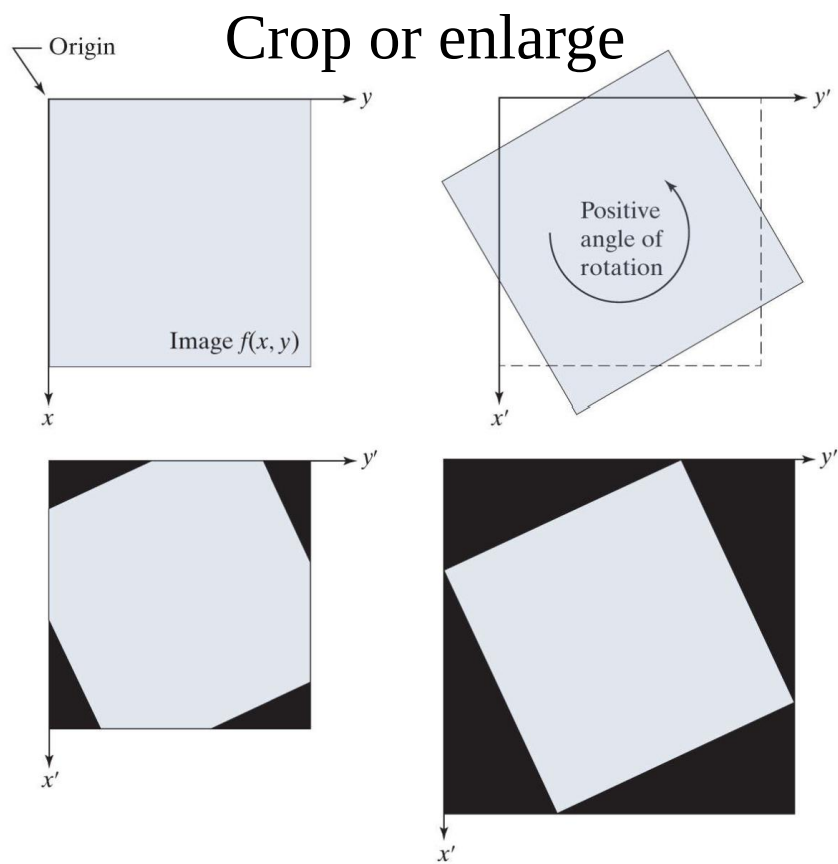
- The overall mapping can be expressed as

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & p \\ 0 & 1 & 0 & q \\ 0 & 0 & 1 & r \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta & 0 & 0 \\ -\sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \omega & 0 & -\sin \omega & 0 \\ 0 & 1 & 0 & 0 \\ \sin \omega & 0 & \cos \omega & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \phi & \sin \phi & 0 \\ 0 & -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

The order of operation matters

# Spatial Transform

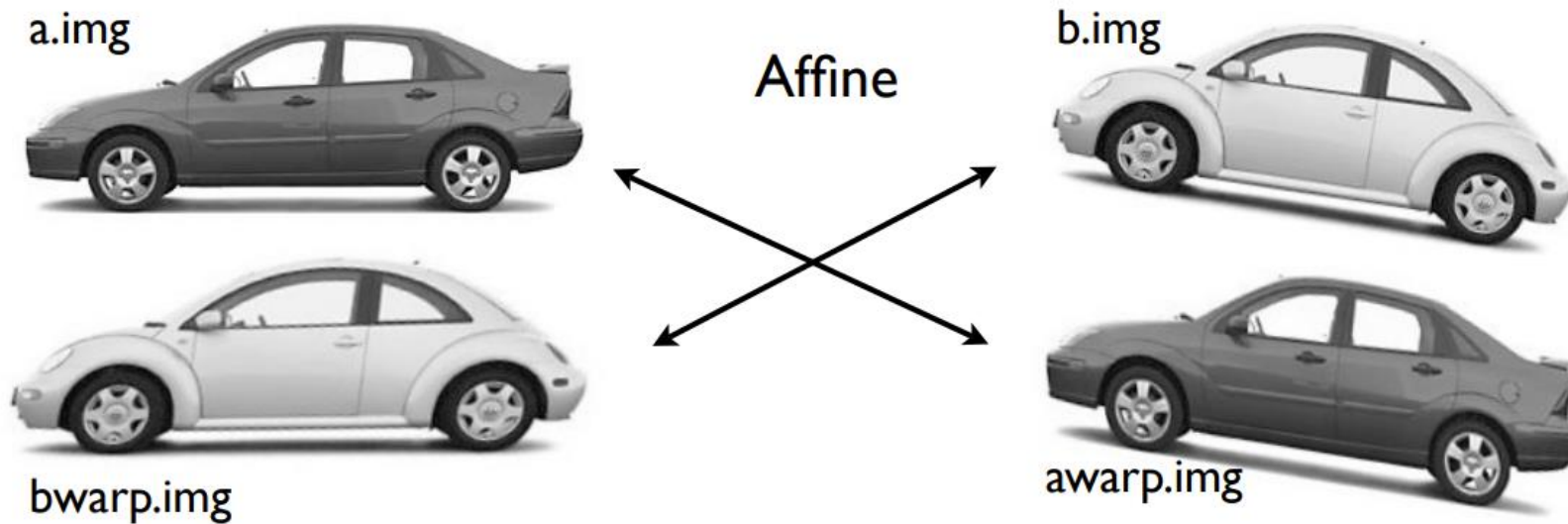
- Interpolation



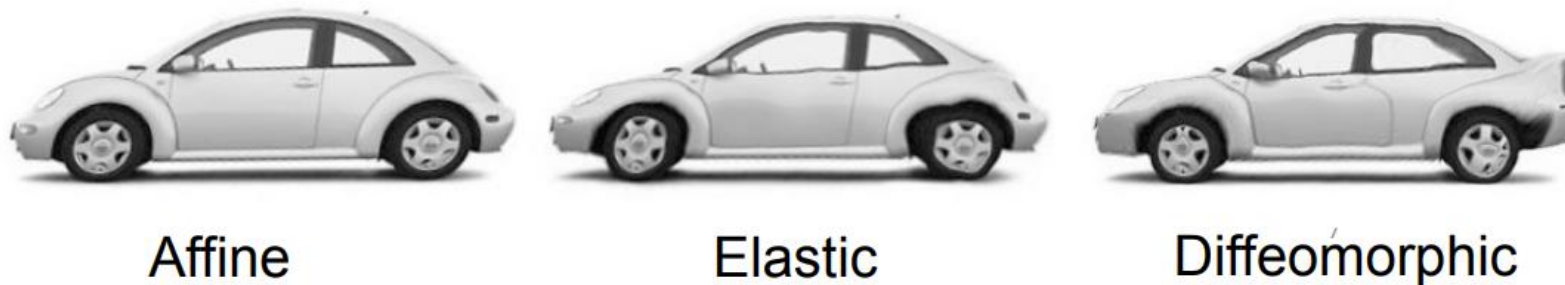


# Non-linear Transform

- Affine transformation (6 degrees of freedom)

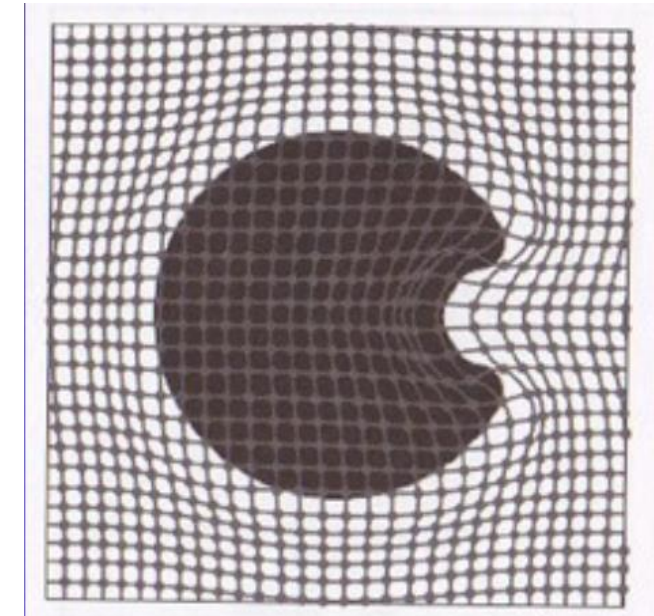
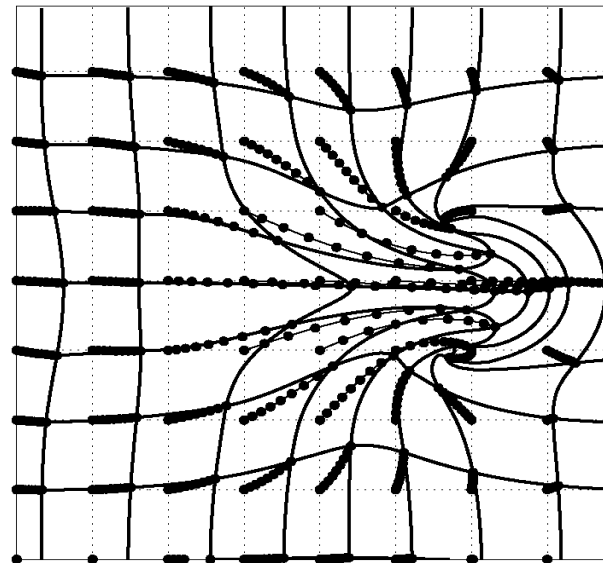
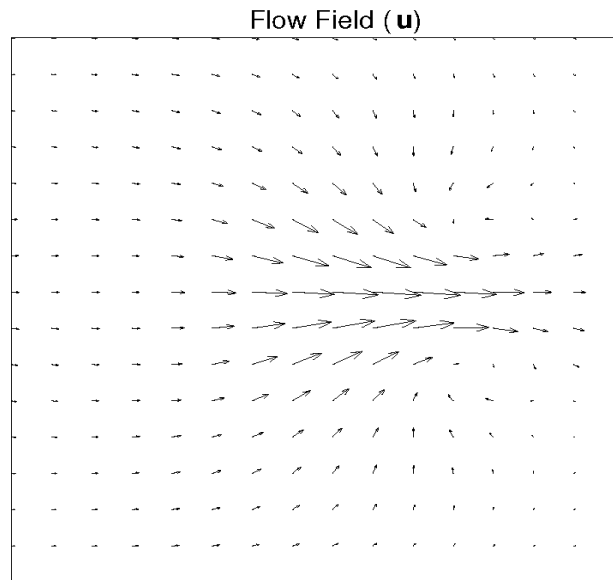


- Increasing degrees of freedom



# Non-linear Transform

- Deformation Field



# Summary

- Mathematical operation
  - Pixel-wised operator and linear operator
  - Arithmetic
    - Add → remove noise / overlap
    - Subtract → difference
    - Multiply → mask
    - Divide → normalize
  - Set and logical operation
- Spatial transform
  - Transformation of the coordinates
  - Intensity interpolation

# Intensity Transformations

Point Processing

Histogram

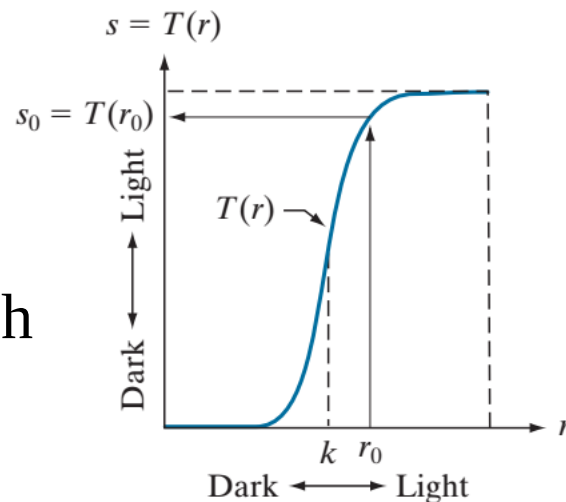
Spatial Filter

# Point processing

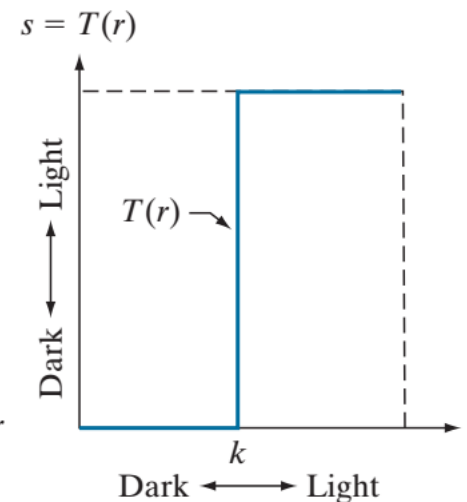
- Point Processing
  - depends only on the gray level at that point.
  - neighborhood is the size 1x1
- $T$  becomes a gray-level (an intensity or mapping)

$$s = T(r)$$

Focus →  
contrast stretch

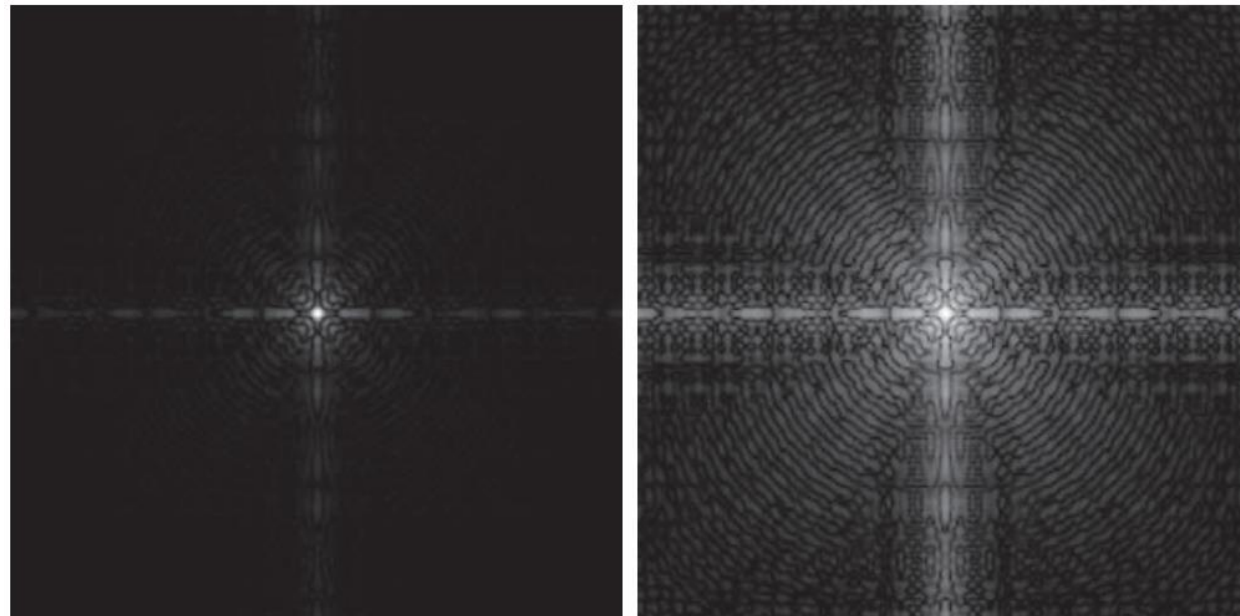
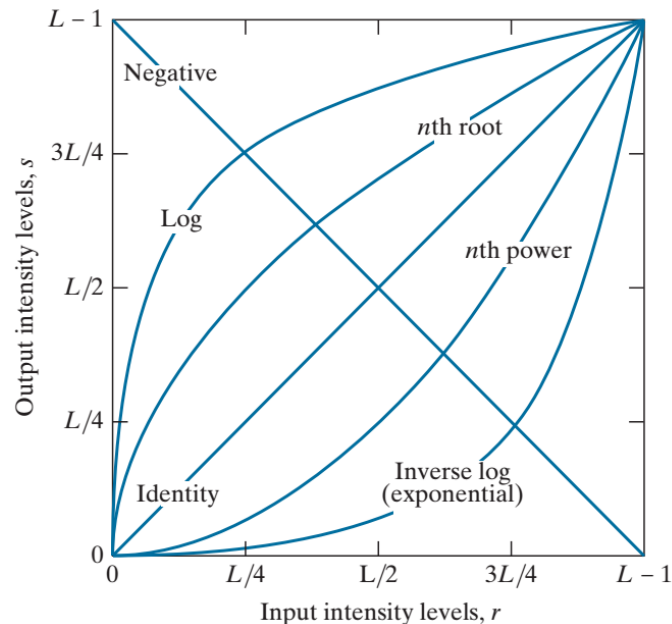


Threshold →  
binary



# Point processing

- Gray Level Transformations
  - linear (negative and identity transformations)  $s = L - r$
  - logarithmic (log and inverse-log transformations)  $s = c \log(1 + r)$
  - power-law (nth power root, gamma transformations)  $s = cr^\gamma$

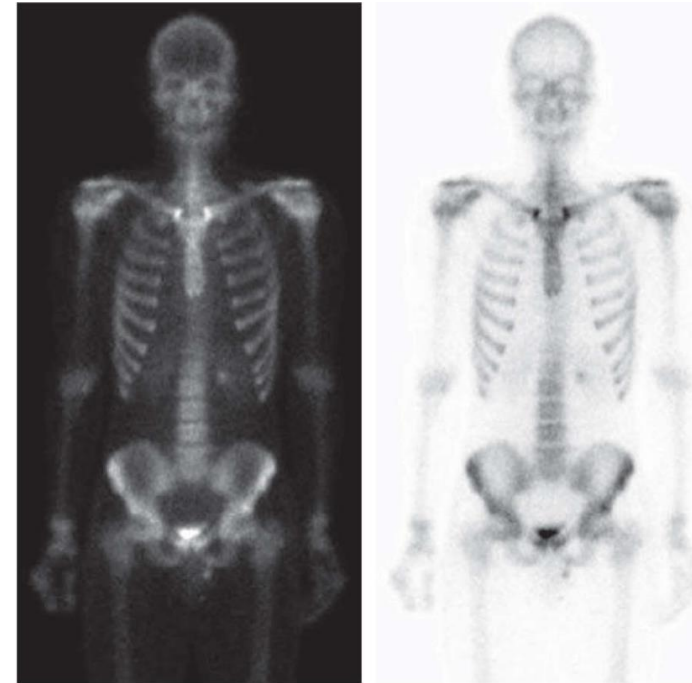
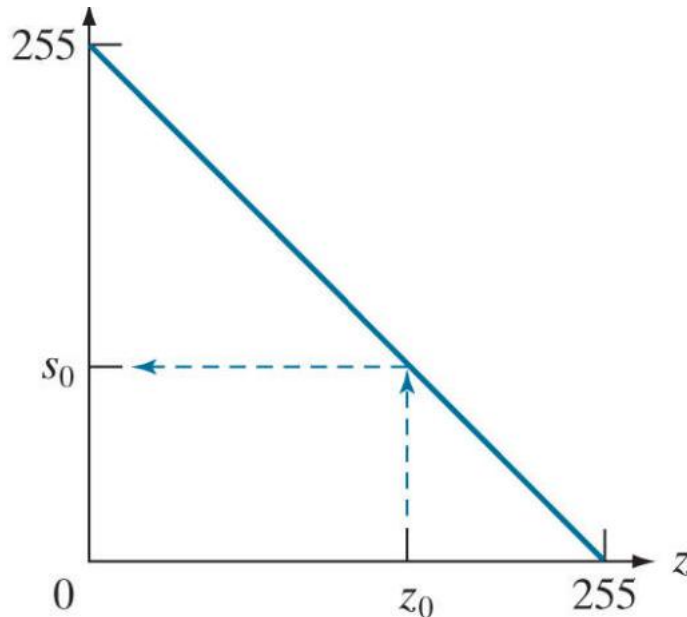


# Point processing

- Negative transform

$$p(x, y) = z \quad s = T(z)$$

$$A^c = \{(x, y, K - z) \mid (x, y, z) \in A\}$$

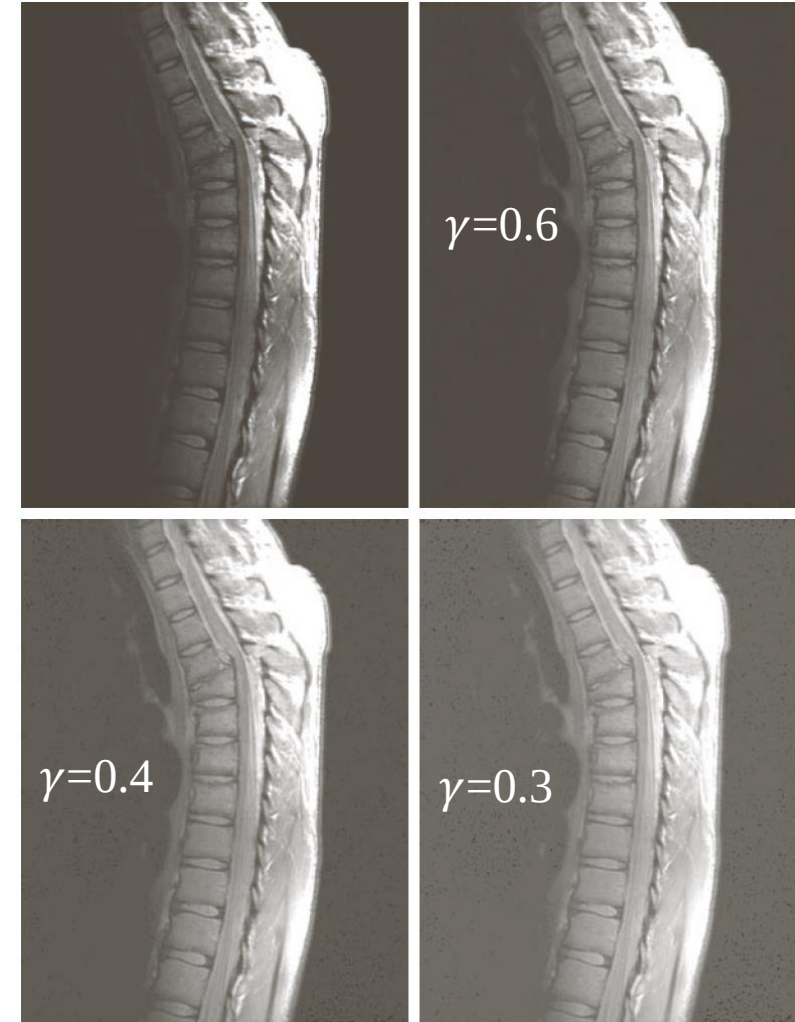
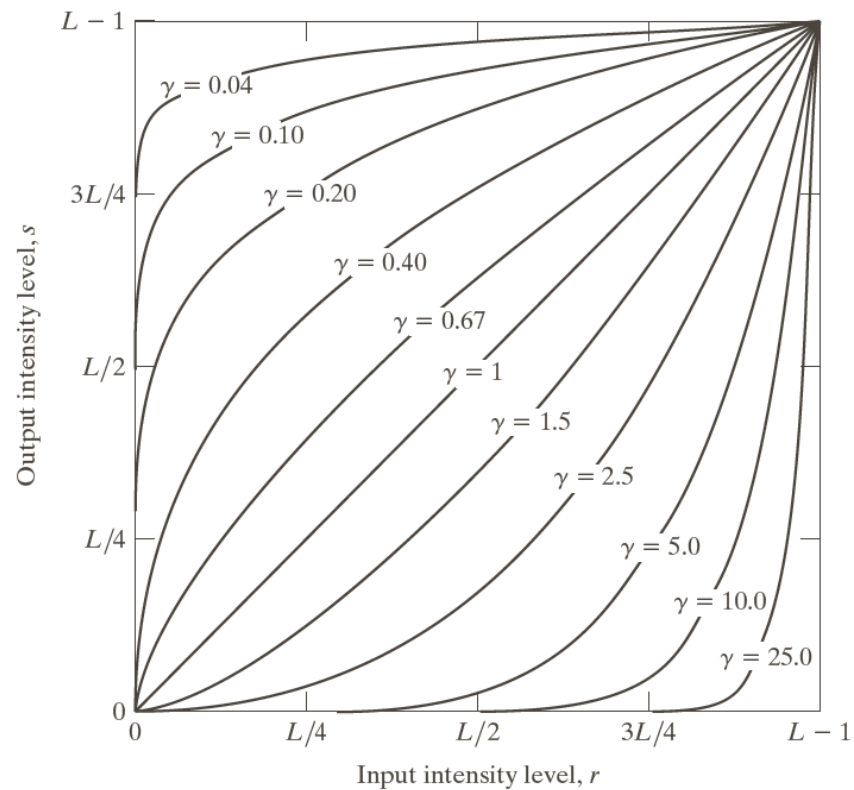


$$s = T(z) = 255 - z$$



# Point processing

- Power-law (Gamma) transform

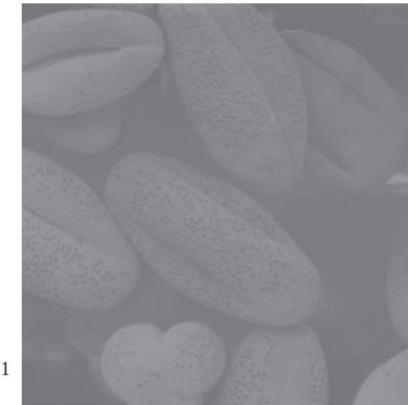
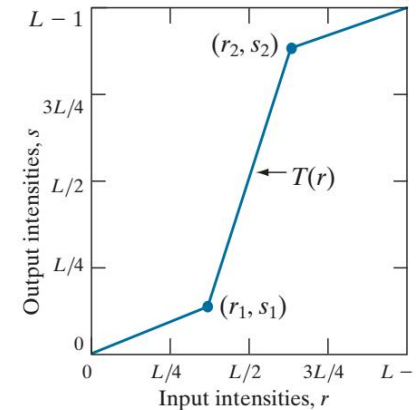




# Point processing

- Piecewise-Linear transformation
  - Arbitrarily complex
  - More user input

Contrast stretching



original

enhanced

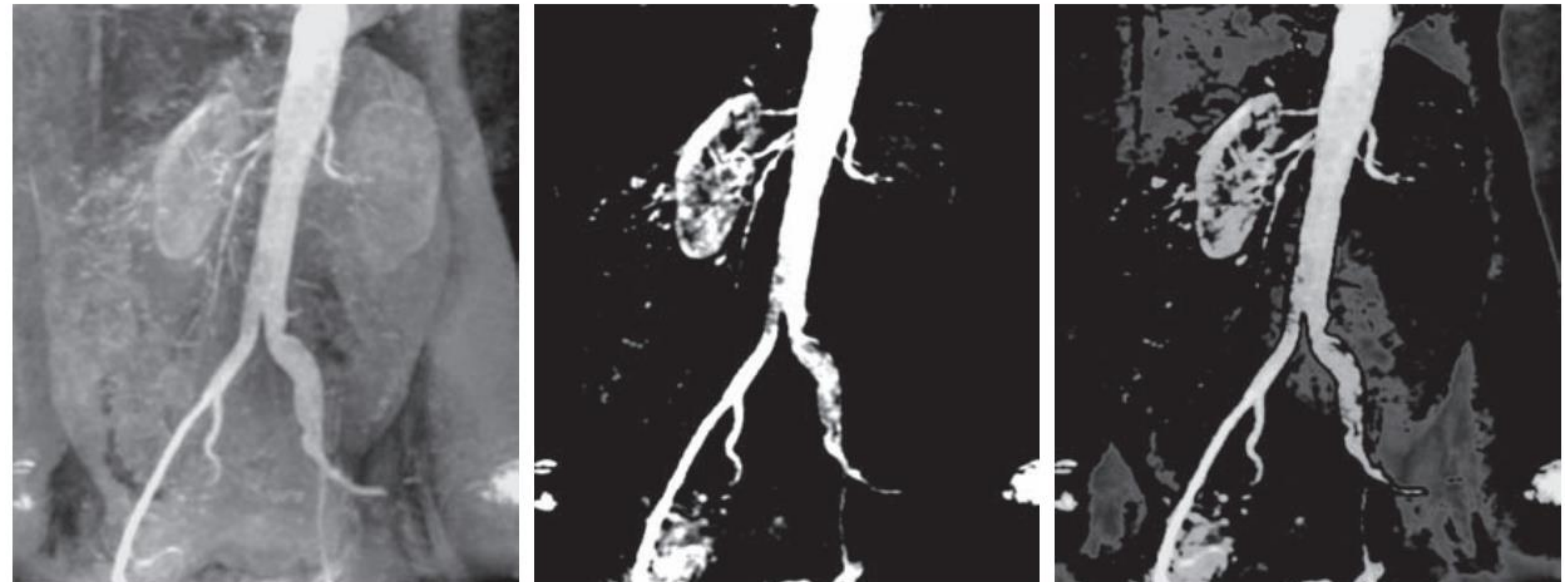
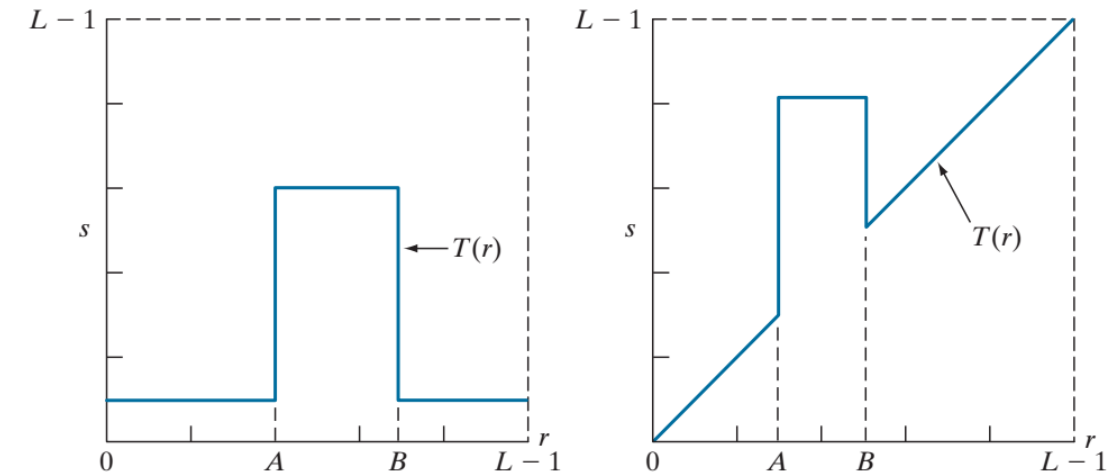


threshold



# Point processing

- Intensity-level slicing
  - Highlighting a specific range of gray levels
  - Remove background
  - Enhance foreground



# Intensity Transformations

Point Processing

Histogram

Spatial Filter

# Histogram

- The number of pixels at each gray level

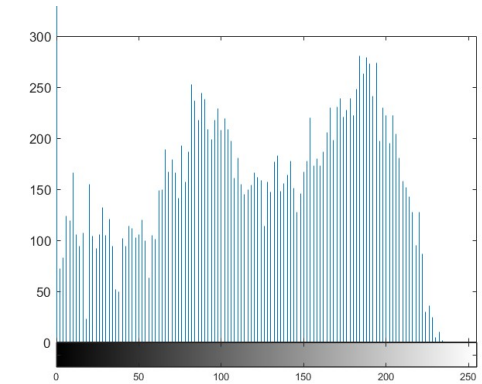
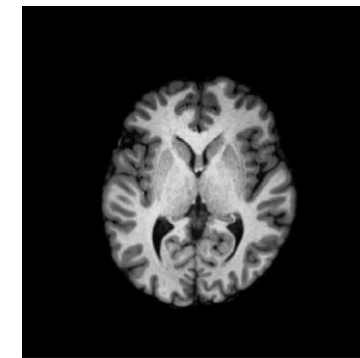
$$h(r_k) = n_k, k \in [0, L - 1]$$

- Normalized histogram

- Probability Density Function

$$p(r_k) = \frac{h(r_k)}{MN}$$

- No spatial information
- Different images could have identical histograms



# Histogram

- Statistics of the histogram

1<sup>st</sup> order  
Mean intensity  
Brightness

$$E = \sum_k^L z_k p(z_k)$$

2<sup>nd</sup> order  
Spread of intensity  
Contrast

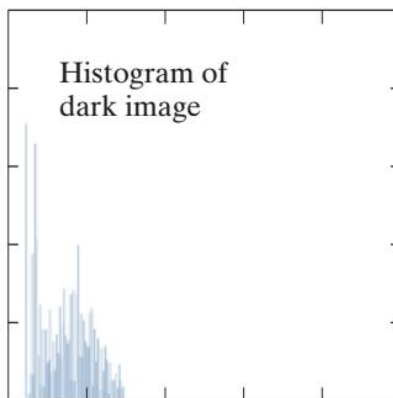
$$std = \sum_k^L (z_k - E)^2 p(z_k)$$

N<sup>th</sup> order

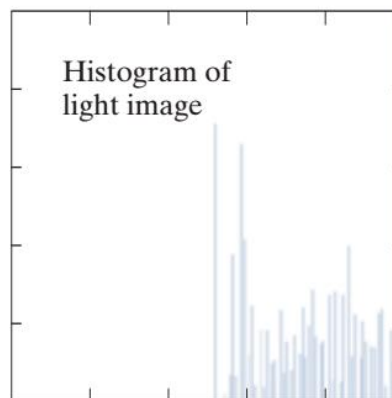
$$\mu_n = \sum_k^L (z_k - E)^n p(z_k)$$

# Histogram

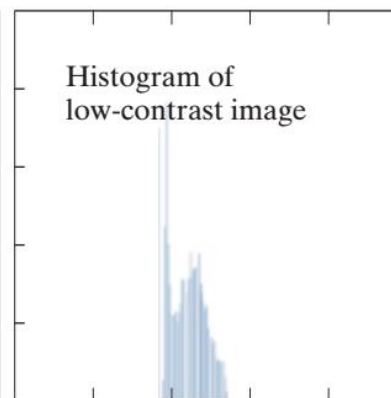
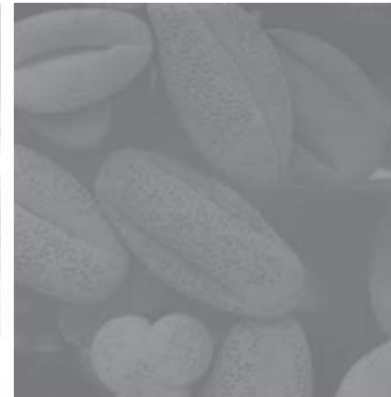
Dark



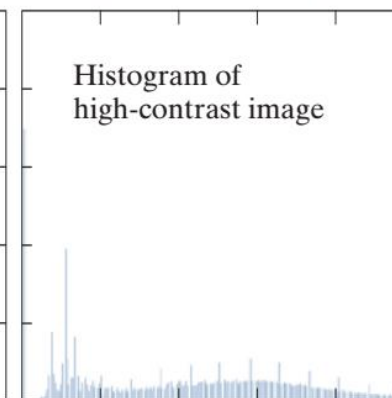
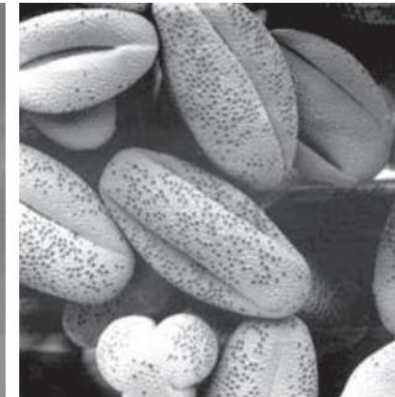
Light



Low contrast



High contrast



# Histogram

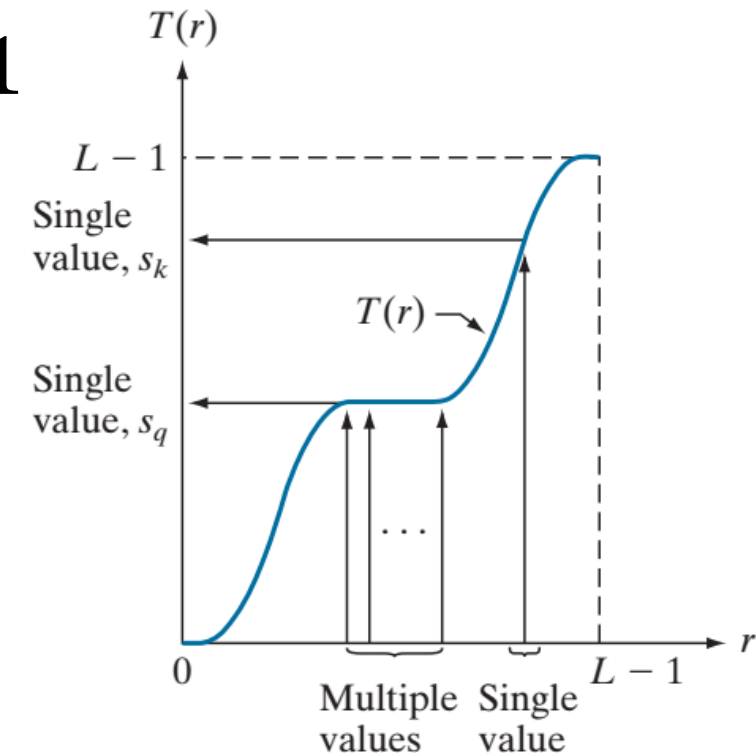
- Forward transform
- Assume: continuous functions

$$s = T(r), \quad 0 \leq r \leq L - 1$$

- $T(r)$  is a non-decreasing function
- $0 \leq T(r) \leq 1$  for  $0 \leq r \leq 1$

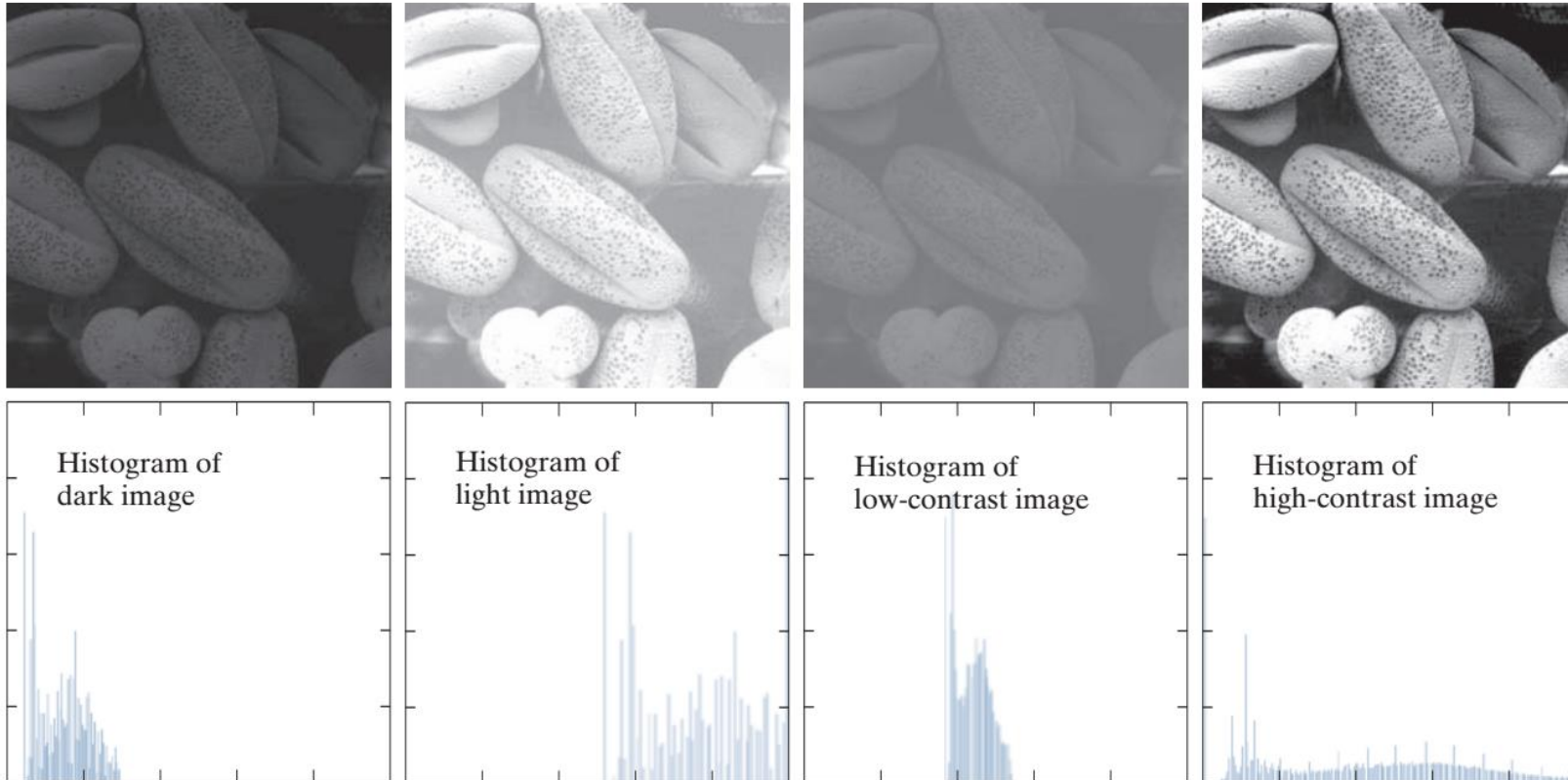
- Inverse transformation from

$$r = T^{-1}(s)$$



# Histogram Equalization

- Goal: design  $s = T(r)$ , so that  $p_s(s) = \frac{1}{L-1}$





# Histogram Equalization

- $p_r$  and  $p_s$ : PDF of the original and converted image

$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right| \quad \text{If } T(r) \text{ is continuous \& differentiable}$$

If we choose  $T(r)$  as

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$

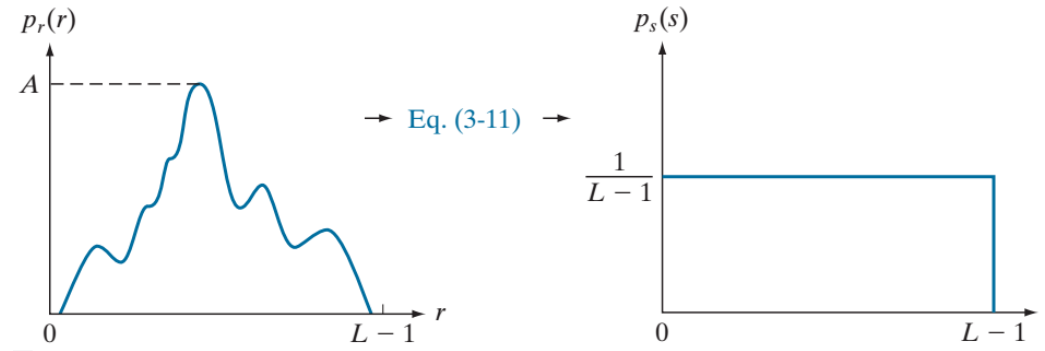
Then

$$\frac{ds}{dr} = \frac{dT(r)}{dr} = (L - 1) \frac{d}{dr} \left[ \int_0^r p_r(w) dw \right] = (L - 1) p_r(r)$$

$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right| = p_r(r) \left| \frac{1}{(L - 1) p_r(r)} \right| = \frac{1}{L - 1}$$

# Histogram Equalization

- Continuous



- Discrete

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$



$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$

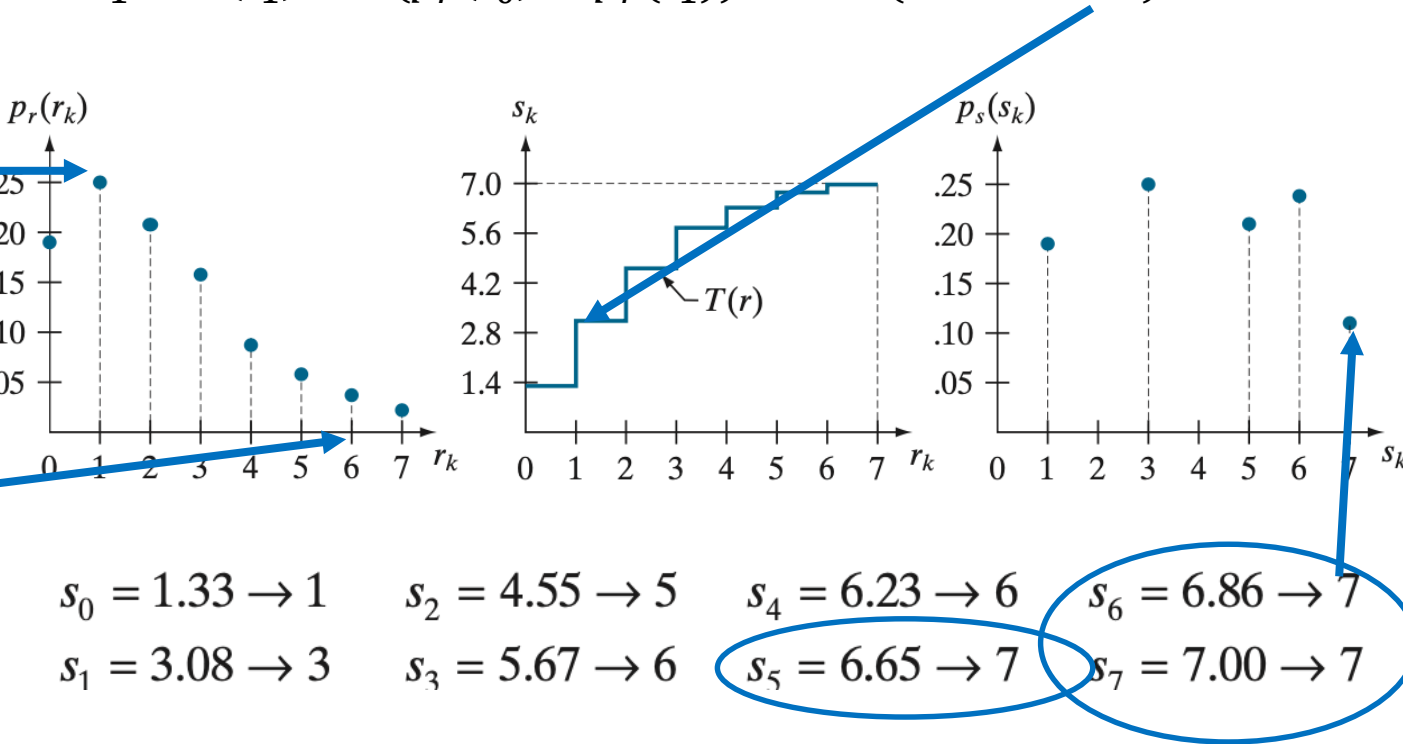
# Example

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$

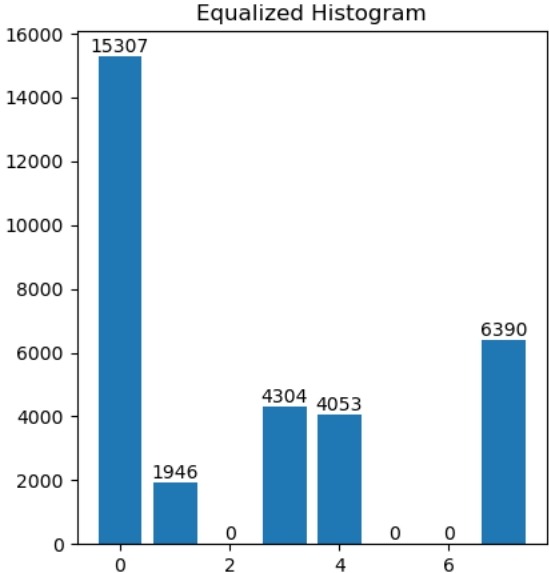
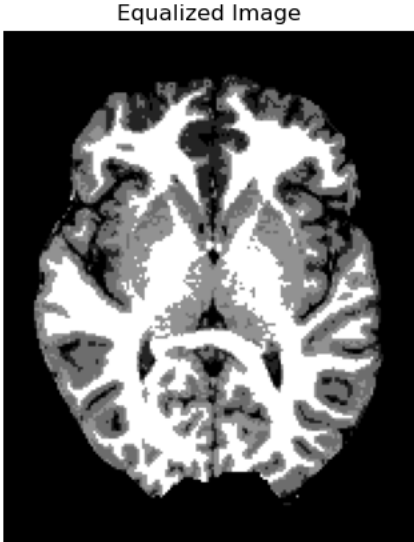
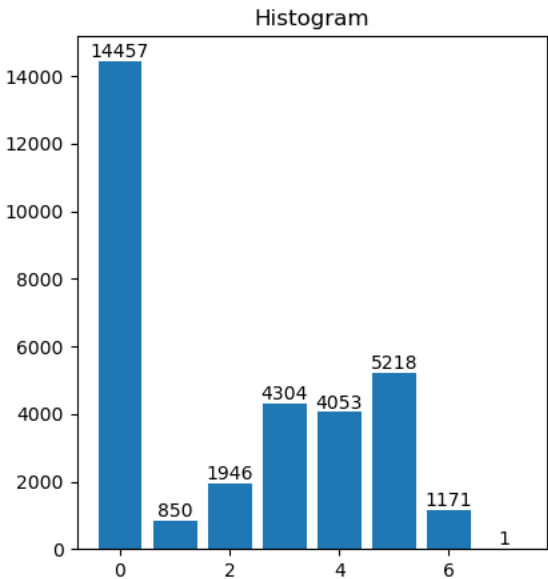
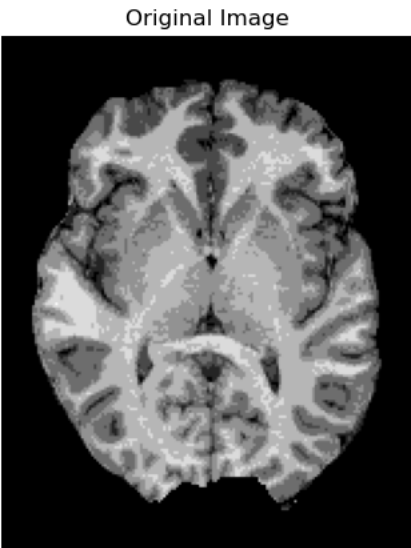
- A 64x64 image Histogram

| $r_k$     | $n_k$ | $p_r(r_k) = n_k / MN$ |
|-----------|-------|-----------------------|
| $r_0 = 0$ | 790   | 0.19                  |
| $r_1 = 1$ | 1023  | 0.25                  |
| $r_2 = 2$ | 850   | 0.21                  |
| $r_3 = 3$ | 656   | 0.16                  |
| $r_4 = 4$ | 329   | 0.08                  |
| $r_5 = 5$ | 245   | 0.06                  |
| $r_6 = 6$ | 122   | 0.03                  |
| $r_7 = 7$ | 81    | 0.02                  |

64x64=4096



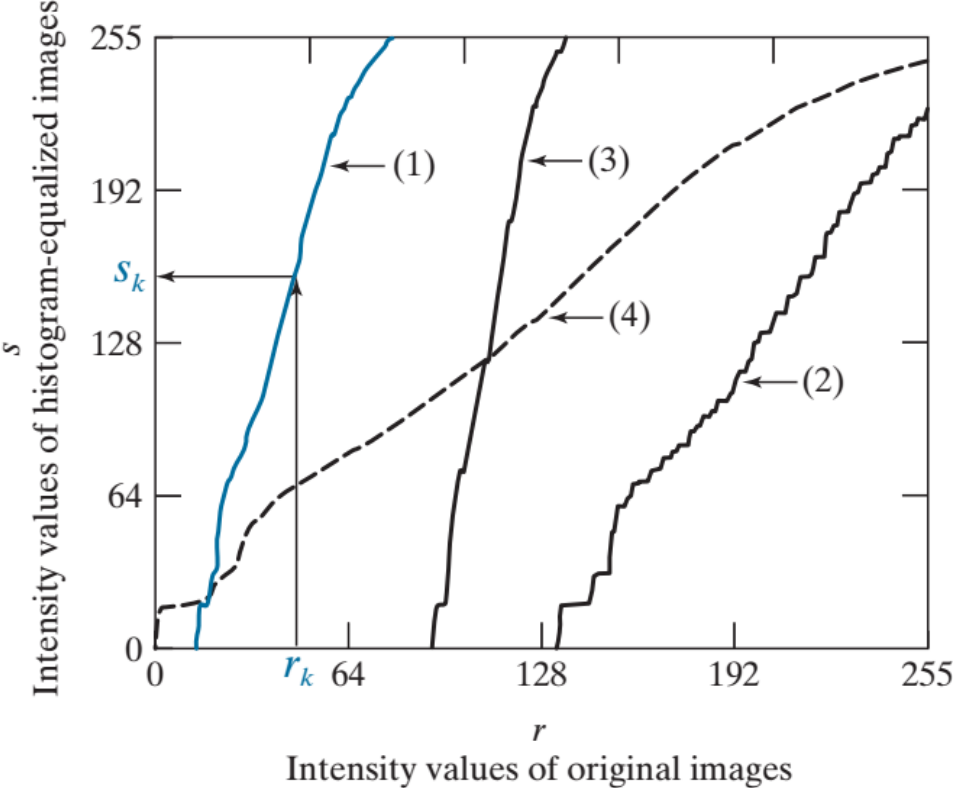
# Example



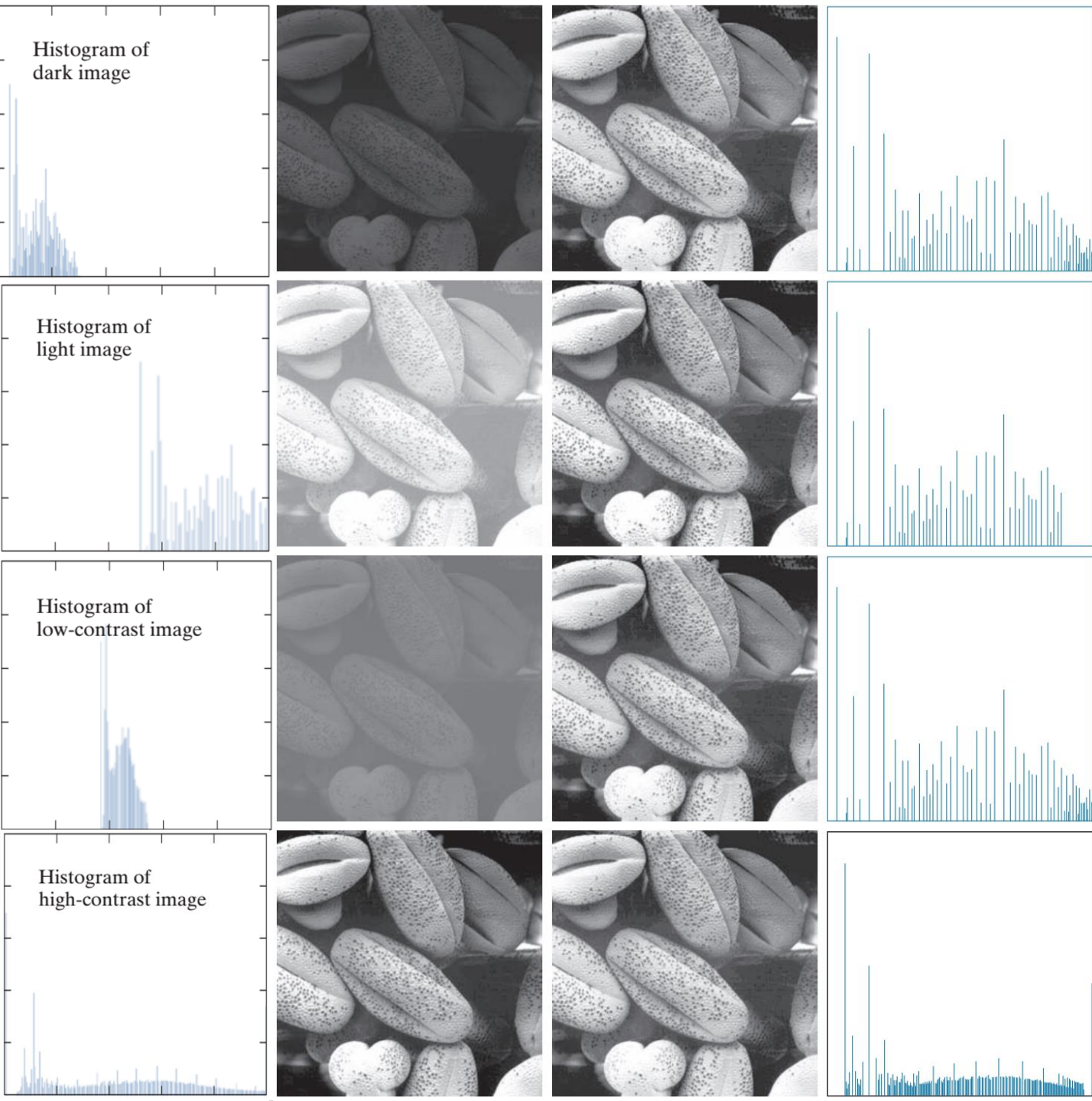
When the background region contains excessive zero-value pixels, histogram equalization is typically computed starting from intensity level 1 (ignoring the zero-value pixels)

| $r_k$             | 0     | 1      | 2      | 3      | 4      | 5      | 6      | 7      |
|-------------------|-------|--------|--------|--------|--------|--------|--------|--------|
| $n_k$             | 14457 | 850    | 1946   | 4304   | 4053   | 5218   | 1171   | 1      |
| Remove Background | 0     | 850    | 1946   | 4304   | 4053   | 5218   | 1171   | 1      |
| $p(n_k)$          | 0     | 0.0485 | 0.1109 | 0.2453 | 0.2310 | 0.2974 | 0.0668 | 0.0001 |
| Cdf               | 0     | 0.0485 | 0.1594 | 0.4047 | 0.6358 | 0.9332 | 0.9999 | 1      |
| (L-1)*Cdf         | 0     | 0.3392 | 1.1157 | 2.8330 | 4.4503 | 6.5323 | 6.9996 | 7      |
| $S_k$             | 0     | 0      | 1      | 3      | 5      | 7      | 7      | 7      |

# Histogram

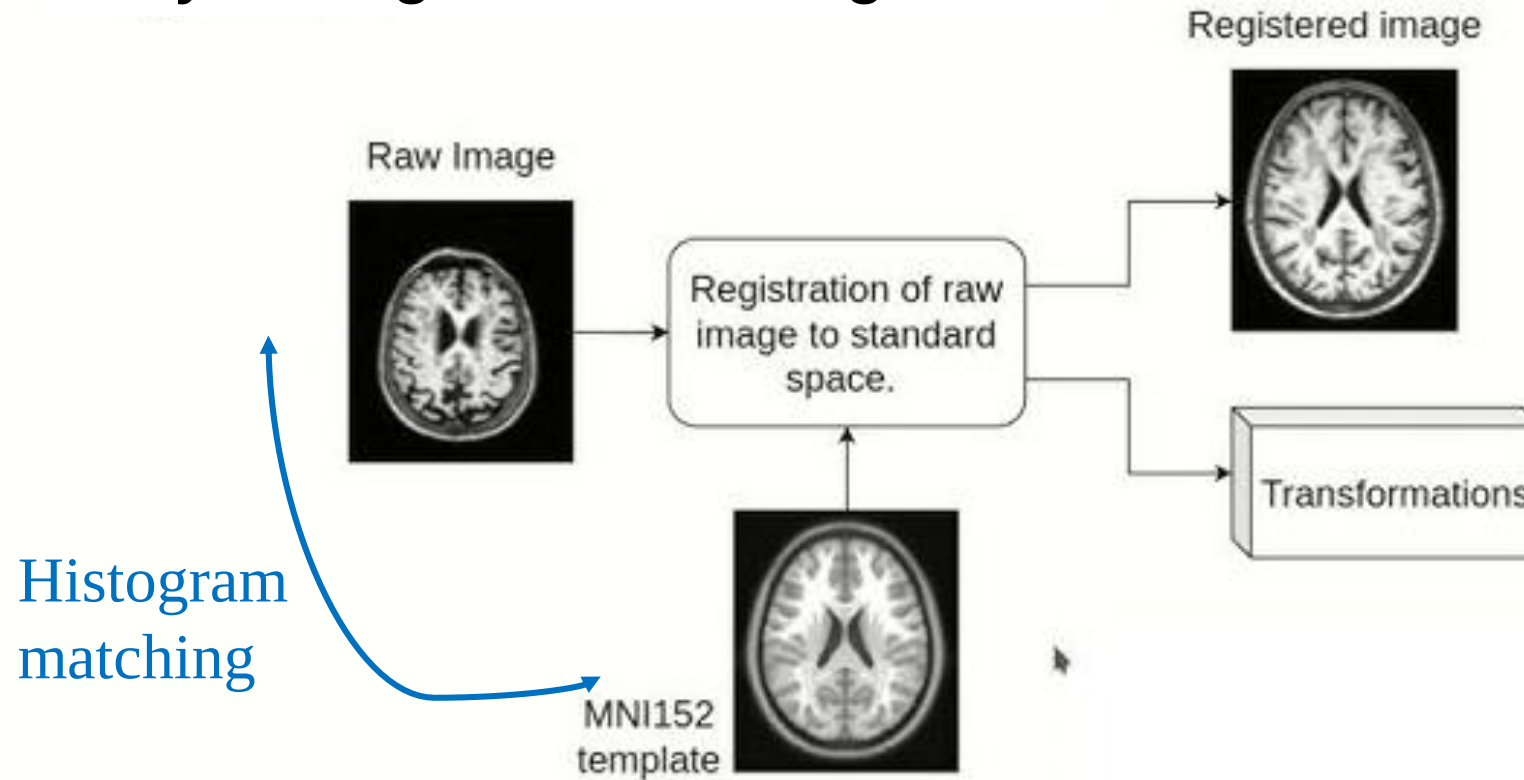


Equalization is not perfect



# Histogram Matching

Why histogram matching?



# Histogram Matching

- Histogram equalization  $\rightarrow$  uniform histogram
- Histogram matching  $\rightarrow$  Specified histogram shape

Define:

|            |                                |                  |
|------------|--------------------------------|------------------|
| $s = T(r)$ | $= (L - 1) \int_0^r p_r(w) dw$ | Input image      |
| $s = G(z)$ | $= (L - 1) \int_0^z p_z(v) dv$ | Target histogram |

$$z = G^{-1}(s) = G^{-1}[T(r)]$$

# Histogram Matching

- Digital

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$



$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$

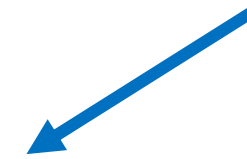
$$s = G(z) = (L - 1) \int_0^z p_z(v) dv$$

$$s_k = G(z_k) = (L - 1) \sum_{i=0}^k p_z(z_i)$$

- Algorithm

- $p_r$  of the input image, round the values  $s_k$
- $G(z_k)$  of the target histogram → Lookup table
- Lookup the table from  $s_k$  to  $z_k$  and match the intensity

No inverse





# Example

- Input histogram

| $r_k$     | $n_k$ | $p_r(r_k) = n_k / MN$ |
|-----------|-------|-----------------------|
| $r_0 = 0$ | 790   | 0.19                  |
| $r_1 = 1$ | 1023  | 0.25                  |
| $r_2 = 2$ | 850   | 0.21                  |
| $r_3 = 3$ | 656   | 0.16                  |
| $r_4 = 4$ | 329   | 0.08                  |
| $r_5 = 5$ | 245   | 0.06                  |
| $r_6 = 6$ | 122   | 0.03                  |
| $r_7 = 7$ | 81    | 0.02                  |

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$

$$s_0 = 1.33 \rightarrow 1 \quad s_2 = 4.55 \rightarrow 5 \quad s_4 = 6.23 \rightarrow 6 \quad s_6 = 6.86 \rightarrow 7$$

$$s_1 = 3.08 \rightarrow 3 \quad s_3 = 5.67 \rightarrow 6 \quad s_5 = 6.65 \rightarrow 7 \quad s_7 = 7.00 \rightarrow 7$$

• Target Histogram

| $z_q$     | Specified<br>$p_z(z_q)$ |
|-----------|-------------------------|
| $z_0 = 0$ | 0.00                    |
| $z_1 = 1$ | 0.00                    |
| $z_2 = 2$ | 0.00                    |
| $z_3 = 3$ | 0.15                    |
| $z_4 = 4$ | 0.20                    |
| $z_5 = 5$ | 0.30                    |
| $z_6 = 6$ | 0.20                    |
| $z_7 = 7$ | 0.15                    |

Example

|                               |                               |
|-------------------------------|-------------------------------|
| $G(z_0) = 0.00 \rightarrow 0$ | $G(z_4) = 2.45 \rightarrow 2$ |
| $G(z_1) = 0.00 \rightarrow 0$ | $G(z_5) = 4.55 \rightarrow 5$ |
| $G(z_2) = 0.00 \rightarrow 0$ | $G(z_6) = 5.95 \rightarrow 6$ |
| $G(z_3) = 1.05 \rightarrow 1$ | $G(z_7) = 7.00 \rightarrow 7$ |

Lookup table

| $z_q$     | $G(z_q)$ |
|-----------|----------|
| $z_0 = 0$ | 0        |
| $z_1 = 1$ | 0        |
| $z_2 = 2$ | 0        |
| $z_3 = 3$ | 1        |
| $z_4 = 4$ | 2        |
| $z_5 = 5$ | 5        |
| $z_6 = 6$ | 6        |
| $z_7 = 7$ | 7        |

|                            |                            |                            |                            |
|----------------------------|----------------------------|----------------------------|----------------------------|
| $s_0 = 1.33 \rightarrow 1$ | $s_2 = 4.55 \rightarrow 5$ | $s_4 = 6.23 \rightarrow 6$ | $s_6 = 6.86 \rightarrow 7$ |
| $s_1 = 3.08 \rightarrow 3$ | $s_3 = 5.67 \rightarrow 6$ | $s_5 = 6.65 \rightarrow 7$ | $s_7 = 7.00 \rightarrow 7$ |

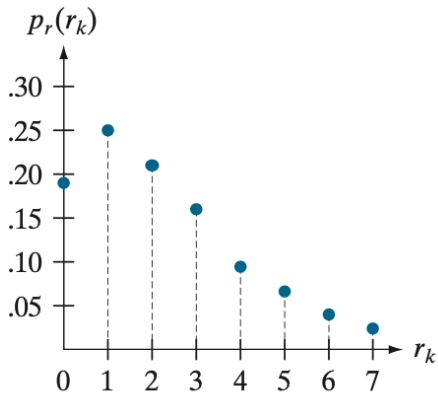
| $s_k$ | $\rightarrow$ | $z_q$ |
|-------|---------------|-------|
| 1     | $\rightarrow$ | 3     |
| 3     | $\rightarrow$ | 4     |
| 5     | $\rightarrow$ | 5     |
| 6     | $\rightarrow$ | 6     |
| 7     | $\rightarrow$ | 7     |

- ①  $s_k = T(r_k)$
- ②  $s_k \rightarrow G(z_q) \rightarrow z_q$

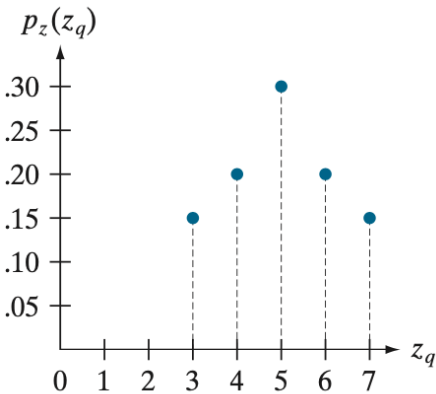
# Example

| $z_q$     | Specified<br>$p_z(z_q)$ | Actual<br>$p_z(z_q)$ |
|-----------|-------------------------|----------------------|
| $z_0 = 0$ | 0.00                    | 0.00                 |
| $z_1 = 1$ | 0.00                    | 0.00                 |
| $z_2 = 2$ | 0.00                    | 0.00                 |
| $z_3 = 3$ | 0.15                    | 0.19                 |
| $z_4 = 4$ | 0.20                    | 0.25                 |
| $z_5 = 5$ | 0.30                    | 0.21                 |
| $z_6 = 6$ | 0.20                    | 0.24                 |
| $z_7 = 7$ | 0.15                    | 0.11                 |

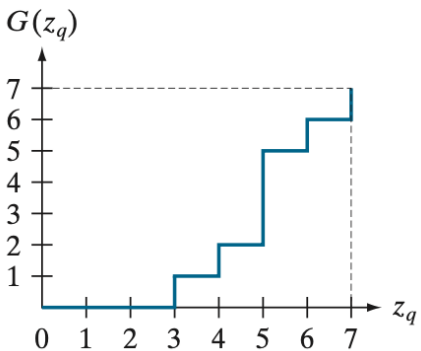
Input  
Histogram



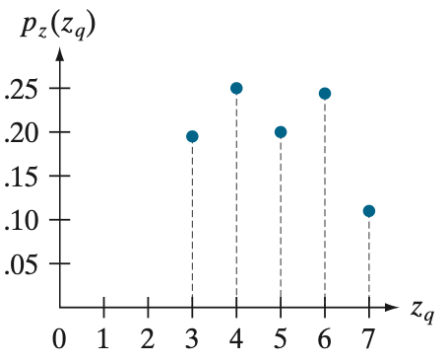
Target  
Histogram



Transform

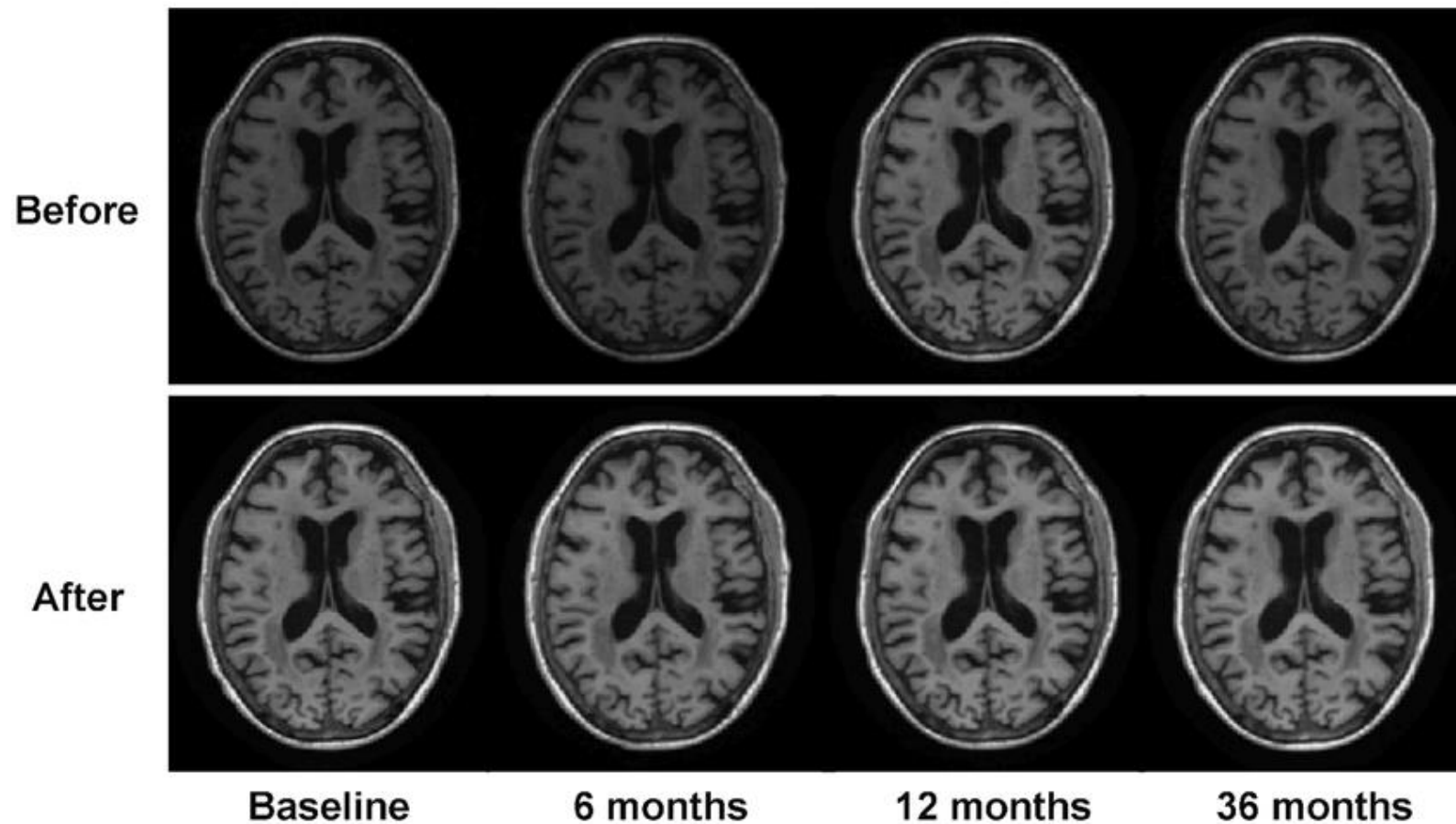


Result  
Histogram



# Example

- N4 correction



# Summary

- Histogram
  - Statistical
  - Related to image: brightness and contrast
  - Histogram transform
- Histogram Equalization
  - To be uniform
- Histogram Matching
  - To be the same as target

# Intensity transformations

Point Processing

Histogram

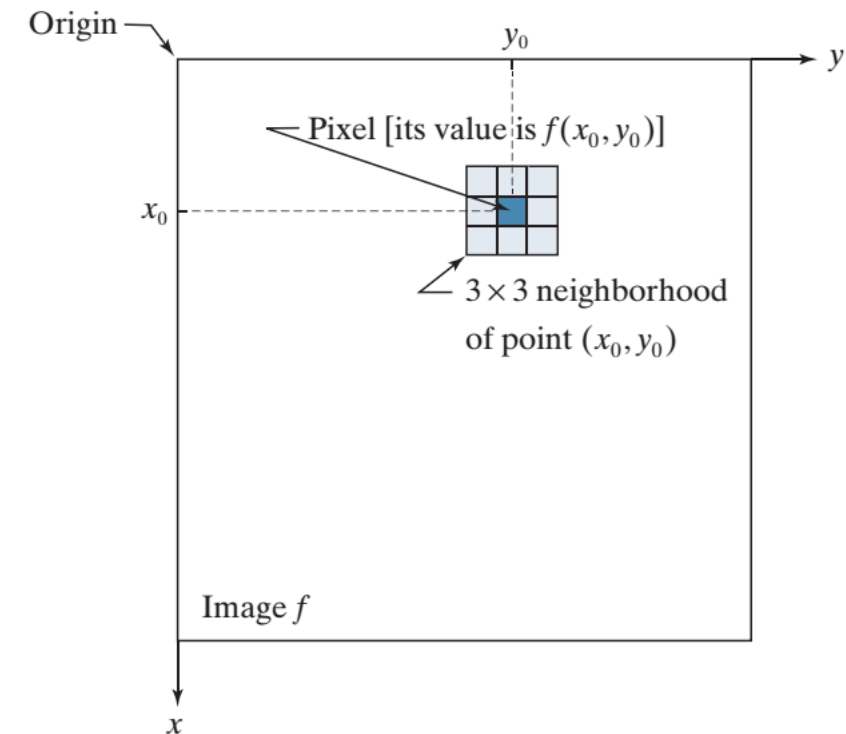
Spatial Filter

# Basic concepts

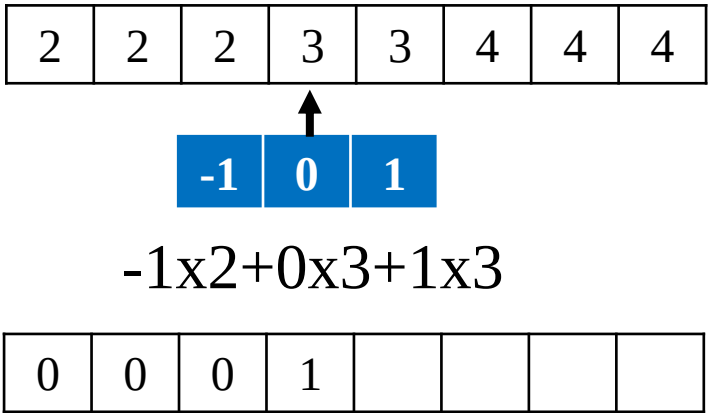
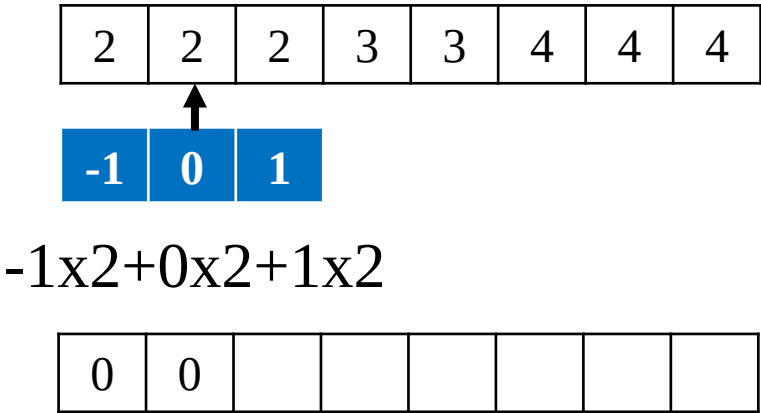
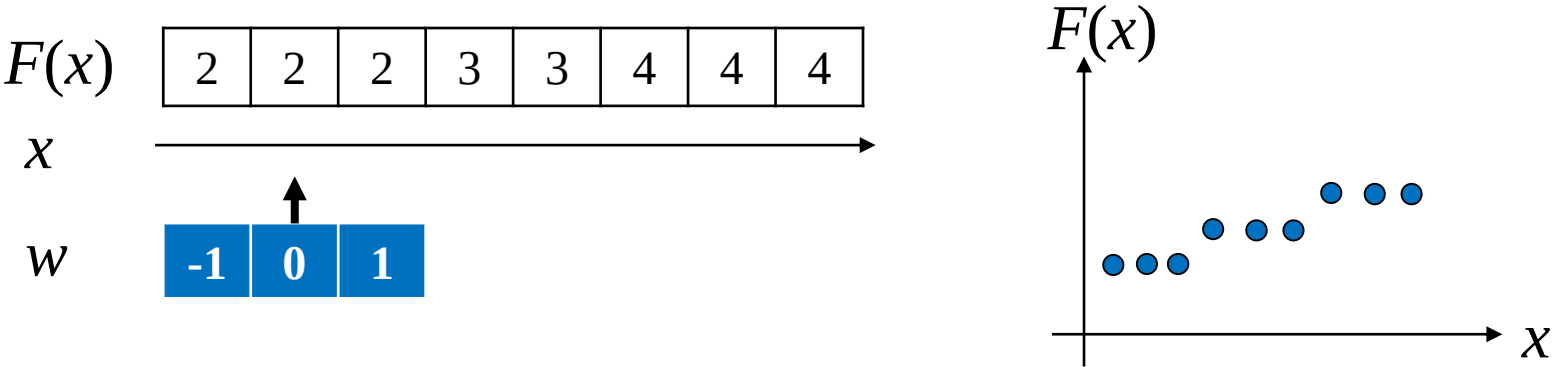
- Spatial Domain
  - Aggregate of pixels composing an image
- Spatial Domain Processes

$$g(x, y) = T(f(x, y))$$

- Principal approach in defining a neighborhood
  - Also called mask, filter, kernel, etc.
  - Values in the mask/filter determine the nature of the process



# Basics of filter

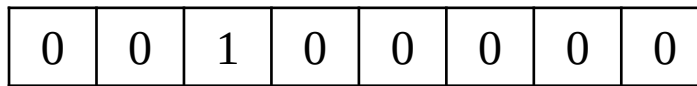




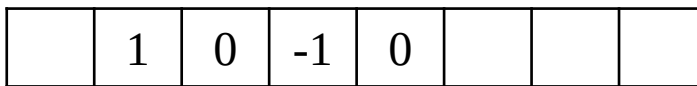
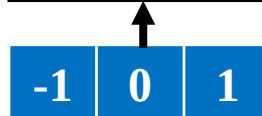
# Basics of filter

## • Correlation

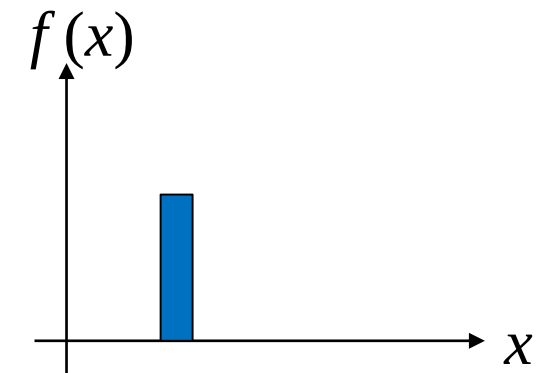
$$g(x) = \sum_{i=-a}^a w(i)f(i+x) \quad x \text{ is the displacement}$$



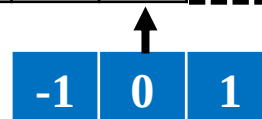
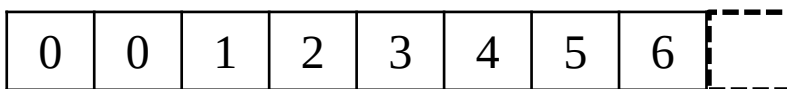
Unit Impulse



Fliped kernel



– Management of edge



- Zero padding
- Mirror

# Basics of filter

- **Convolution:** kernel flipped

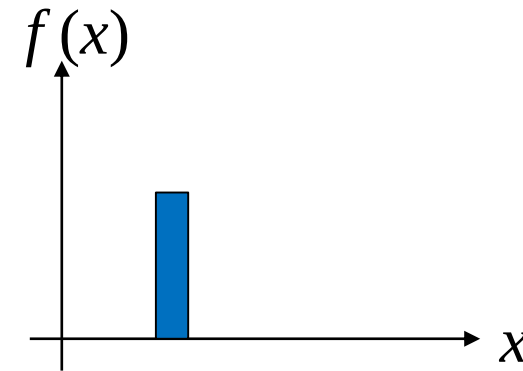
|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
|---|---|---|---|---|---|---|---|

Unit Impulse

|   |   |    |
|---|---|----|
| 1 | 0 | -1 |
|---|---|----|

|  |    |   |   |   |  |  |  |
|--|----|---|---|---|--|--|--|
|  | -1 | 0 | 1 | 0 |  |  |  |
|--|----|---|---|---|--|--|--|

Original kernel



- Same as correlation with a symmetric kernel

# Basics of filter

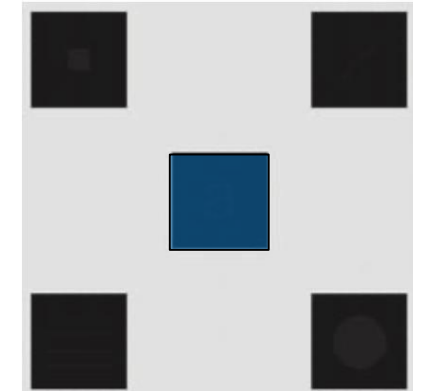
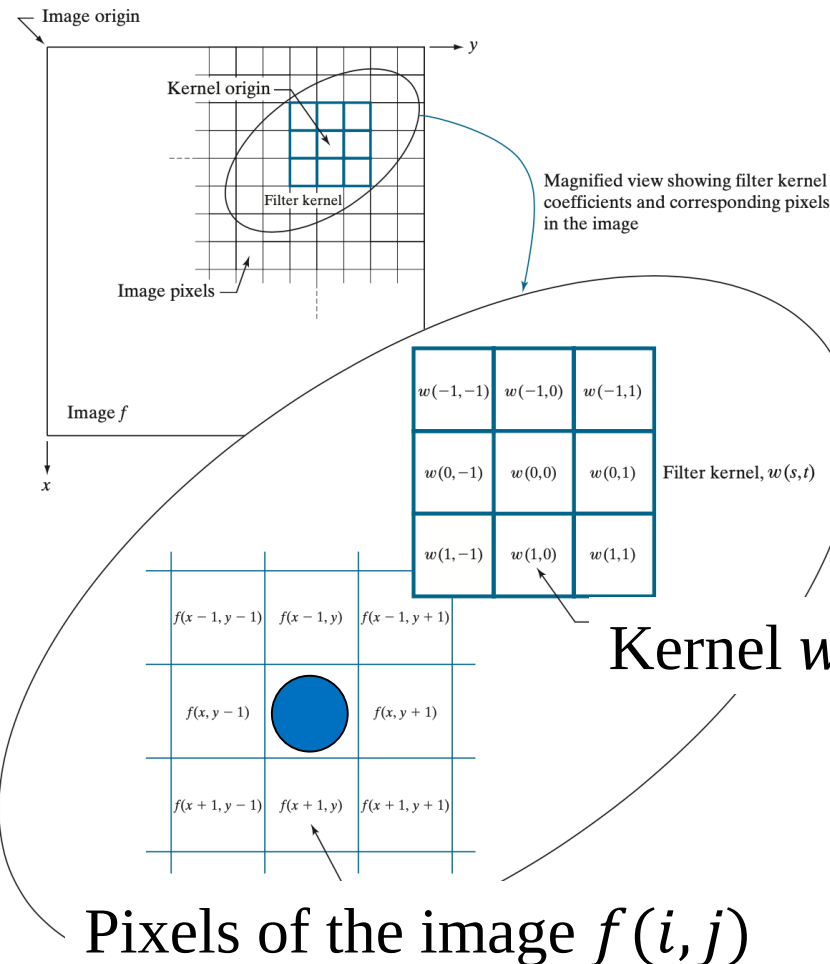
- Filter, mask, kernel, template, window ...



- Coefficients

|    |    |
|----|----|
| 1  | 1  |
| -1 | -1 |

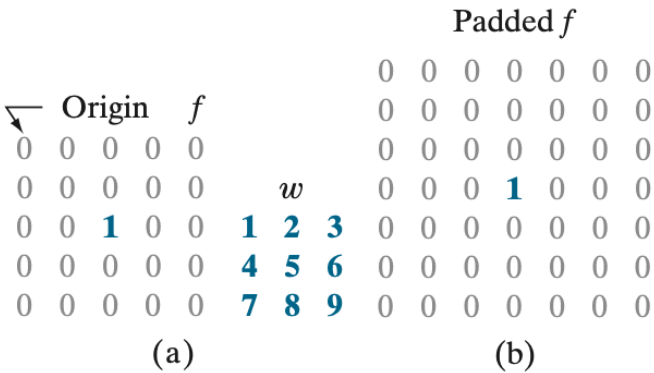
$$g(x, y) = \sum \sum w(i, j) f(i, j)$$



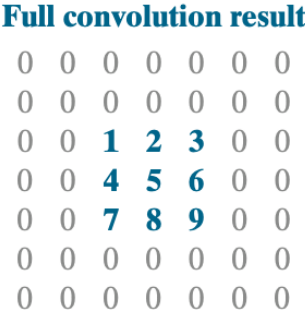
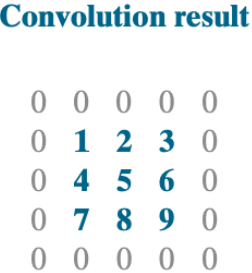
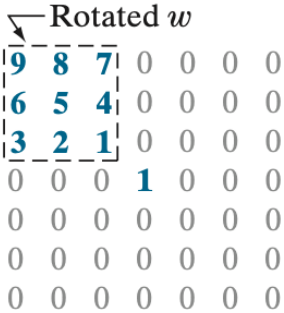
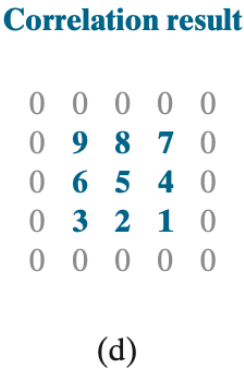
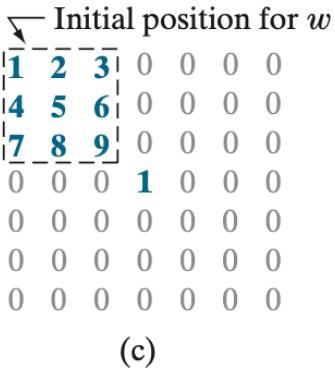
Size is odd number  
3x3, 5x5, ...



# Basics of filter



Correlation  
VS  
Convolution



## Basics of filter

- $f \star g = g \star f$
- $f \star (g + h) = f \star g + f \star h$
- $f \star (g \star h) = (f \star g) \star h$

- Separatable

$$\mathbf{c} \mathbf{r}^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = w$$

- Normalized

# Intensity Transformations

- Pixel-wise
  - Linear, log, gamma
- Histogram (Probability)
  - Histogram Equalization
  - Histogram Match
- Regional
  - Convolution
  - Filter

# Homework 1

## 2D Spatial Affine Transform

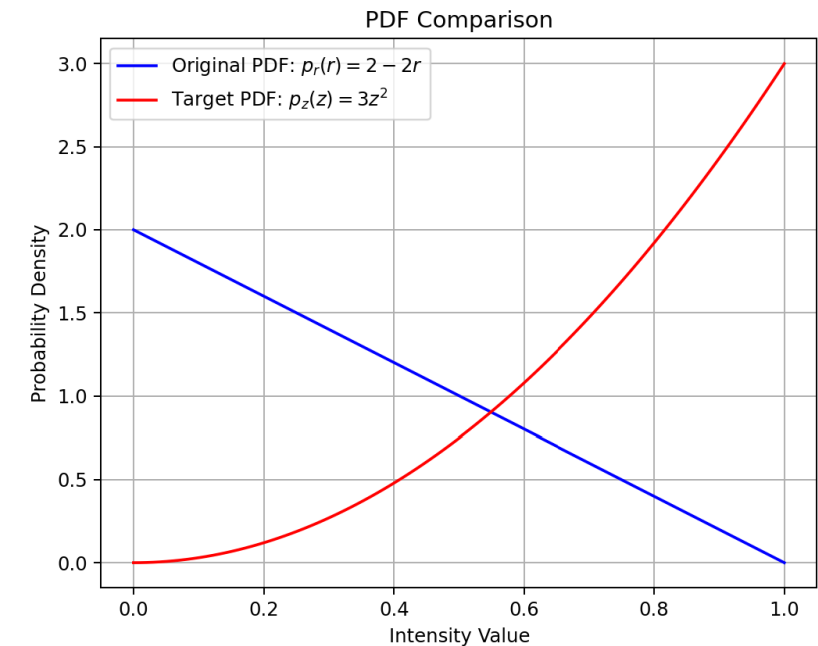
- Triangle A has vertices at  $A1(0,0)$ ,  $A2(2,0)$ ,  $A3(0,1)$ .
- Triangle B has vertices at  $B1(2,1)$ ,  $B2(2,5)$ ,  $B3(3,1)$ .

### Tasks:

- Design a sequence of 3x3 affine transformation matrices (homogeneous coordinates).
- 1. Define the matrices for Scaling (S), Rotation (R), and Translation (T)  
(Order:  $S \rightarrow R \rightarrow T$ ).
- 2. Calculate the overall transformation matrix  $M = T * R * S$ .

# Homework 2

- Given two random variables  $R$  and  $Z$  with the following probability density functions (PDFs):  
 $R$  has a PDF  $p_r(r)$ ,  $p_r(r) = 2 - 2r$   
 $Z$  has a PDF  $p_z(z)$ ,  $p_z(z) = 3z^2$
- Find the intensity transformation function  $z = T(r)$  that maps the original intensity  $r$  to the target intensity  $z$ , such that the transformed original image matches the target PDF  $p_z(z)$





# Homework 3

Discrete Histogram Equalization & Matching, Given a grayscale image ( $L=8$ , intensity levels 0–7) with 100 pixels.

Intensity distribution ( $n_k$ ) for  $k=0$  to 7: [40, 20, 15, 10, 5, 5, 3, 2]

Target probability distribution ( $p_z$ ) for  $z=0$  to 7: [0, 0.07, 0.10, 0.26, 0.28, 0.15, 0.11, 0.03]

- 1. Perform Histogram Equalization on the original image to find the discrete mapping values  $s_k$ .
- 2. Compute the transformation function  $G(z)$  for the target histogram.
- 3. Perform Histogram Matching to map original intensity  $r_k$  to target intensity  $z_q$ .
  - In case of a tie, choose the smaller  $z_q$ .